

Gut microbiome in metabolic dysfunction-associated steatotic liver disease and associated hepatocellular carcinoma

Harry Cheuk-Hay Lau^{1,2}, Xiang Zhang^{1,2} & Jun Yu¹✉

Abstract

Metabolic dysfunction-associated steatotic liver disease (MASLD) is the most prevalent chronic liver disease worldwide, affecting billions of the global population. It can gradually progress to more severe diseases, including steatohepatitis, cirrhosis and hepatocellular carcinoma. Studies have highlighted the importance of the gut microbiome in the pathogenesis and progression of MASLD. On the other hand, increasing evidence has revealed the clinical potential of targeting the gut microbiome to treat MASLD. In this Review, we summarize gut microbial alterations in MASLD, metabolic dysfunction-associated steatohepatitis and hepatocellular carcinoma. The mechanisms by which a dysregulated gut–liver axis contributes to disease progression are also described, including intestinal barrier dysfunction, chronic inflammation, and altered metabolic pathways (for example, bile acids) and microbial-derived metabolites (for example, short-chain fatty acids, tryptophan derivatives and endogenous ethanol). In addition, we discuss the clinical implications of utilizing the gut microbiome as a diagnostic biomarker and the therapeutic approaches to treat MASLD and related diseases such as faecal microbiota transplantation, probiotics and engineered bacteria, prebiotics and postbiotics, microbial-derived metabolites, antimicrobials and bacteriophages. Finally, we discuss current challenges in basic and translational research on the microbiome in MASLD and propose future directions to drive progress in this field.

Sections

Introduction

Gut microbiome in MASLD, MASLD-related cirrhosis and MASLD-related HCC

Mechanisms of dysregulated gut–liver axis in MASLD and MASLD-related HCC

Clinical applications of the gut microbiome in MASLD and MASLD-related HCC

Current challenges and future directions

Conclusions

¹Institute of Digestive Disease, Department of Medicine and Therapeutics, State Key Laboratory of Digestive Disease, Li Ka Shing Institute of Health Sciences, CUHK Shenzhen Research Institute, The Chinese University of Hong Kong, Hong Kong SAR, China. ²These authors contributed equally: Harry Cheuk-Hay Lau, Xiang Zhang.

✉e-mail: junyu@cuhk.edu.hk

Key points

- Progression from metabolic dysfunction-associated steatotic liver disease (MASLD) to steatohepatitis, fibrosis, cirrhosis and, eventually, hepatocellular carcinoma is closely associated with gut microbial dysbiosis and a dysfunctional gut–liver axis.
- As MASLD progresses, the gut microbiome exhibits enriched Proteobacteria, *Clostridium* and *Streptococcus* as well as depleted beneficial bacteria including *Eubacterium* and *Faecalibacterium*.
- A dysregulated gut–liver axis as well as microbial dysbiosis promote MASLD progression by impairing the intestinal barrier, altering microbial-mediated metabolic pathways and metabolites, and contributing to chronic inflammation in the liver.
- Gut microorganisms and metabolites can be utilized as diagnostic biomarkers of MASLD, which have been shown to stratify individuals predisposed to an increased risk of disease progression with high accuracy.
- Owing to its pivotal role, emerging studies have developed various therapeutic approaches that target the gut microbiome to treat MASLD, including faecal microbiota transplantation, probiotics, prebiotics and postbiotics, antimicrobials, microbial-derived metabolites and bacteriophages.
- Research into the microbiome in MASLD is challenged by the lack of mechanistic understanding and consistency across studies, and unclear duration, dosage and efficacy have limited the clinical application of microbiome-targeting interventions.

Introduction

The human intestines and liver are in constant crosstalk to form a ‘gut–liver’ axis. This bidirectional link has two major routes of communication (Fig. 1). First is enterohepatic circulation, in which primary bile acids, cholesterol, immunoglobulin A (IgA) and antimicrobial molecules are produced by the liver and directed to the proximal small intestine via the biliary tract. Bile acids are essential regulators of emulsification and absorption of dietary fats and cholesterol and they are also involved in controlling unrestricted overgrowth of gut bacteria^{1,2}. After being released from the liver, primary bile acids are deconjugated in the small intestine, reabsorbed into the portal circulation and transferred back to the liver for detoxification. Second is intestinal permeability, which selectively enables bile acids (both primary bile acids (cholic acid and chenodeoxycholic acid) and secondary bile acids (deoxycholic acid (DCA) and lithocholic acid)), microbial metabolites, dietary nutrients (for example, amino acids and carbohydrates) and pathogen-associated molecular patterns (PAMPs) to pass through the intestinal barrier and enter the liver through the portal vein. The intestinal barrier is also the first defence that prevents gut microorganisms and potentially harmful molecules from leaving the gut. In normal conditions, this gut–liver axis is fundamental to maintaining intestinal homeostasis but it is highly disrupted in metabolic dysfunction-associated steatotic liver disease (MASLD)³. MASLD is the most common chronic liver disease worldwide⁴. Additionally, 20% of patients with simple steatosis gradually develop metabolic

dysfunction-associated steatohepatitis (MASH), and MASH can then further progress to cirrhosis and eventually hepatocellular carcinoma (HCC)^{5,6} (Box 1). In this Review, we describe gut microbial alterations in MASLD and MASLD-related diseases, such as MASLD-related cirrhosis and MASLD-related HCC, with a focus on the mechanisms underlying how a dysregulated gut–liver axis contributes to disease progression. We then discuss potential approaches that target the gut microbiome to benefit patients with MASLD and propose research directions to further drive progress in this field.

Gut microbiome in MASLD, MASLD-related cirrhosis and MASLD-related HCC

Gut microbial alterations in simple steatosis, MASH and MASLD-related fibrosis

Gut microbial dysbiosis, which refers to dysregulated composition and function of the gut microbiome, is linked to the pathogenesis of MASLD (Fig. 1). Large studies have investigated alterations of the gut microbiome in simple steatosis using 16S rRNA gene sequencing. Compared with healthy individuals, both adult^{7–9} and paediatric¹⁰ patients with simple steatosis or MASH seem to have enriched Proteobacteria (including *Escherichia*) and depleted Firmicutes. However, there are large discrepancies in the abundance of Firmicutes across studies (Table 1). In patients with MASLD or MASH, although *Clostridium*, *Lactobacillus* and *Streptococcus* are consistently enriched and *Coprococcus*, *Eubacterium* and *Faecalibacterium* are depleted (Fig. 1), alterations of other Firmicutes genera such as *Blautia*, *Oscillibacter*, *Roseburia* and *Ruminococcus* remain inconsistent^{7,9–12}.

Whole-genome shotgun metagenomic sequencing has become the mainstay of microbiome investigation as it can characterize microbial taxa at species and strain levels with higher taxonomic resolution than 16S rRNA gene sequencing. In 2017, Loomba et al. conducted the first metagenomic study on patients with MASLD and identified that patients from the USA with MASLD-related fibrosis (stage 3–4 fibrosis) have a significantly lower abundance of *Ruminococcus obeum* and *Eubacterium rectale* compared with patients with MASH (stage 0–2 fibrosis)¹³. Consistently, another study in Europe also reported the depletion of *R. obeum* and *E. rectale* in patients with MASLD, together with decreased abundance of Firmicutes (for example, *Lachnospiraceae*, *Coprococcus catus* and other *Eubacterium* species)¹⁴. Other microorganisms that are consistently associated with MASLD or MASH across human studies include enriched *Escherichia coli*, *Bacteroides* species and *Clostridium* species as well as depleted *Faecalibacterium prausnitzii*^{13–16} (Fig. 1). Interestingly, some adult^{7,14} and paediatric¹⁷ studies reported the enrichment of several well-known probiotics, including *Bifidobacterium*, *Lactobacillus* and *Akkermansia*, in patients with MASLD. Although much evidence has indicated that probiotic consumption has a protective effect on MASLD development^{18,19}, it has been reported that flagellins on certain lactic acid bacteria might trigger a Toll-like receptor (TLR)-mediated immune response in the intestinal epithelium²⁰. Nevertheless, the functional role of these probiotic gut commensals in MASLD progression is yet to be determined.

Most cases of MASLD are linked to obesity, but MASLD also occurs in individuals without obesity with a global prevalence of 3–30%²¹. Although MASLD without obesity has a similar clinical and histopathological presentation to MASLD with obesity²², human studies reported that the gut microbiome is significantly different between patients with MASLD with and without obesity^{23,24}. Lower abundances of *Ruminococcus*, *Lactobacillus*, *Faecalibacterium* and *Eubacterium* were observed in patients with MASLD without obesity from Brazil²³

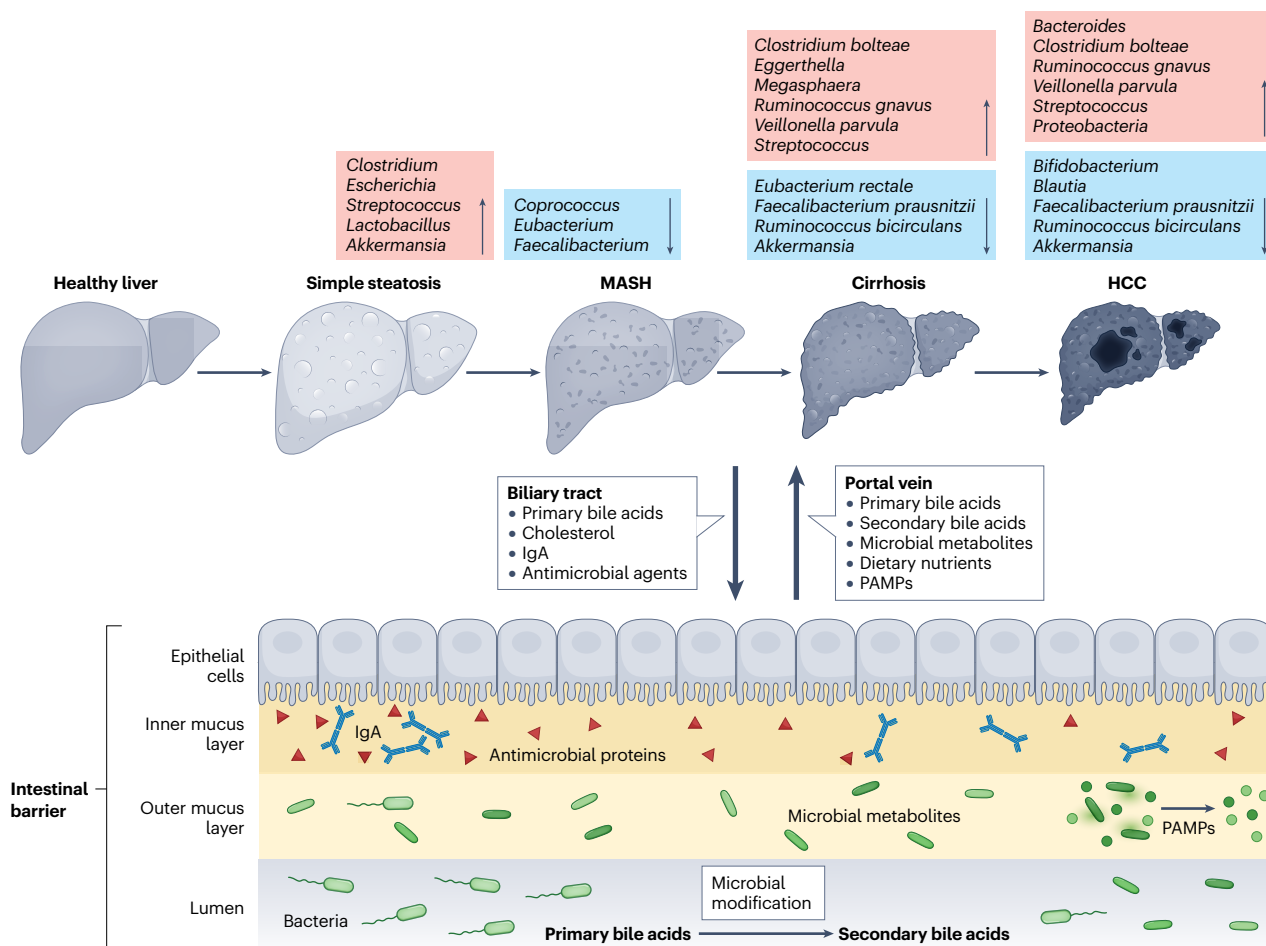


Fig. 1 | Gut–liver axis and gut microbiome alterations during MASLD progression. The gut–liver axis is a bidirectional connection that mainly involves the biliary tract and portal vein. Once primary bile acids, cholesterol, immunoglobulin A (IgA) and antimicrobial molecules are generated by the liver, they are released to the small intestine via the biliary tract. Primary bile acids (for example, cholic acid and chenodeoxycholic acid) are deconjugated into secondary bile acids (for example, deoxycholic acid and lithocholic acid) via gut microorganism-mediated modification (mainly 7 α -dehydroxylation)⁹⁸. On the other hand, bile acids, microbial metabolites, dietary nutrients and pathogen-associated molecular patterns (PAMPs) traverse the intestinal barrier and are transferred back to the liver through the portal circulation. The intestinal barrier is composed of intestinal epithelial cells, an inner mucus layer and outer mucosal

layers. The progression from simple steatosis, metabolic dysfunction-associated steatohepatitis (MASH), to cirrhosis and to hepatocellular carcinoma (HCC) is associated with different compositional changes in the gut microbiome. Gut microbiome alterations along metabolic dysfunction-associated steatotic liver disease (MASLD) progression are featured with enriched Proteobacteria, *Escherichia*, *Clostridium* and *Streptococcus*, which are correlated with high levels of inflammation. On the contrary, the abundance of beneficial gut commensals, including Firmicutes, *Eubacterium* and *Faecalibacterium*, is decreased in MASLD, implying a reduced production of protective and anti-inflammatory metabolites. Several species have varied alterations as MASLD progresses (Table 1) such as *Akkermansia*, which is increased in MASH but decreased in MASLD-related cirrhosis and MASLD-related HCC.

or Japan²⁴ compared with patients with MASLD with obesity. Moreover, gut microbial alterations in patients from Korea with MASLD without obesity are markedly correlated with fibrosis severity, whereas such a correlation is insignificant in obesity-associated MASLD¹². *Klebsiella pneumoniae*, a pathogenic bacterium known to cause pneumonia and bloodstream infection in humans, is enriched in patients with MASLD without obesity with stage 2–3 fibrosis¹². By contrast, the abundances of beneficial gut commensals, including *Ruminococcus*, *E. rectale* and *F. prausnitzii*, are higher in patients with MASLD without obesity and without fibrosis¹². In general, MASLD with obesity and MASLD without obesity have similar pathogenic mechanisms in terms of insulin resistance, triglyceride accumulation and lipotoxicity, whereas patients

without obesity tend to have higher amounts of visceral adipose tissue^{25–27}. However, it is important to note that the unique mechanisms that underlie MASLD without obesity remain poorly understood. Future investigations are required to investigate whether the gut microbiome could contribute to the development of MASLD without obesity.

Gut microbial alterations in MASLD-related cirrhosis

In 2014, an early metagenomic study identified enrichment of oral microorganisms, including multiple *Veillonella* species (for example, *Veillonella parvula* and *Veillonella atypica*) and *Streptococcus* species (for example, *Streptococcus salivarius* and *Streptococcus anginosus*), in the gut microbiome of Chinese patients with cirrhosis²⁸.

Box 1 | Global epidemiology and progression of MASLD and MASLD-related conditions

Metabolic dysfunction-associated steatotic liver disease (MASLD; formerly known as nonalcoholic fatty liver disease)²⁴⁴ is the most common chronic liver disease worldwide, with a global prevalence of 25%⁴. Metabolic risk factors, particularly type 2 diabetes mellitus, are tightly linked to the onset of MASLD²⁴⁵. Latin America is the only region that has a prevalence of MASLD above 40% (44.37%)²⁴⁶. The prevalence of MASLD in most regions is approximately 30–35% (for example, Middle East and North Africa (36.53%), South Asia (33.83%), South-East Asia (33.07%) and North America (31.20%)) while Western Europe has the lowest prevalence around the world (25.10%)²⁴⁶. Owing to the continuous increase in the number of individuals with metabolic disorders, especially in developed countries where overnutrition and unhealthy lifestyles are more common, it is speculated that the prevalence of MASLD could continue to rise and affect over 30% of the global population²⁴⁶.

MASLD includes a spectrum of liver diseases. Metabolic dysfunction-associated steatotic liver is the prerequisite of several more severe liver pathologies. In particular, 20% of patients with metabolic dysfunction-associated steatotic liver or simple steatosis develop metabolic dysfunction-associated steatohepatitis (MASH), whereas MASH can gradually advance to fibrosis, cirrhosis and, eventually, the end-stage liver malignancy hepatocellular carcinoma (HCC)^{5,6}. Remarkably, MASLD is the fastest-growing aetiology of HCC; the global incidence of MASLD-related HCC has increased by fourfold in the past decade²⁴⁷. Despite being a global health burden, there is currently only one approved drug for MASH treatment (resmetirom)²⁴⁸. Meanwhile, several other promising compounds, including semaglutide²⁴⁹, pegozafermin²⁵⁰, efruxifermin²⁵¹ and lanifibranor²⁵², are currently in phase III clinical trials and have shown great potential to treat MASH. In light of a projected rise in disease incidence, it is imperative to develop novel and effective therapeutic strategies to treat MASLD and MASLD-related conditions.

Notably, although this work did not consider the aetiology of cirrhosis, similar findings were reported in later studies, in which oral microorganisms (especially *Streptococcus*) were markedly enriched in European patients²⁹ and patients from the USA³⁰ with MASLD-related cirrhosis. The enrichment of oral microorganisms was further confirmed in subsequent metagenomic studies as evidenced by increased abundance of *V. parvula*, *V. atypica*, *S. salivarius* and *Ruminococcus gnavus* in the gut microbiome of patients with MASLD-related cirrhosis from the USA³¹ or Australia³² (Fig. 1). The consistent enrichment of oral microorganisms across human studies implies that these species might be translocated and colonized into the intestines to promote disease progression. Meanwhile, patients with MASLD-related cirrhosis tend to have lower abundance of gut commensal probiotics such as *Akkermansia*²⁹, *Bacillus* and *Lactococcus*³⁰. In particular, *E. rectale* and *F. prausnitzii*, two prominent butyrate-producing bacteria with anti-inflammatory functions^{33,34}, are significantly depleted in patients with cirrhosis²⁸ or MASLD-related cirrhosis³¹ as well as in patients with MASLD or MASH (Table 1). Owing to consistent observations of their depletion from simple steatosis to more severe cirrhosis, the reduction

of these two probiotic species might be a hallmark feature in the gut microbiome of MASLD.

Gut microbial alterations in MASLD-related HCC

To date, only a small number of studies have profiled the gut microbiome in patients with MASLD-related HCC. Similarly to its precursor diseases, the gut microbiome of MASLD-related HCC is enriched with oral microorganisms, particularly *Streptococcus*, and is accompanied by the depletion of the probiotics *Bifidobacterium* and *Akkermansia*²⁹. In 2021, Behary et al. conducted one of the largest metagenomic studies on patients with MASLD-related HCC and MASLD-related cirrhosis³². Bacterial species with increased abundance in both MASLD-related HCC and MASLD-related cirrhosis include *Bacteroides xylanisolvens*, *R. gnavus* and *Clostridium bolteae*, whereas *Bacteroides caecimuris* and *V. parvula* are enriched only in MASLD-related HCC. Interestingly, all five of these MASLD-related HCC-enriched species are producers of short-chain fatty acids (SCFAs) with immunomodulatory activities^{35–37}. As SCFAs are generally thought to be beneficial to health through their antitumour functions, these results imply that SCFAs might have a debatable role in MASLD-related HCC development (discussed later). The enrichment of specific SCFA-producing bacteria and immunomodulatory species, in addition to the depletion of beneficial gut commensals, collectively reshape the tumour microenvironment to favour immunosuppression and tumorigenesis in MASLD-related HCC. Nonetheless, it is noteworthy that microbial signatures of MASLD-related HCC were mainly identified from a single human cohort without external validation^{29,32}.

In general, study variance has long been a critical issue in microbiome research. Although some microbial species are consistently altered in multiple studies, others have been inconsistent (Table 1). As the gut microbiome is readily influenced by environmental factors (such as age, sex and geographical region), careful interpretation when comparing findings across studies is necessary. For example, most microbiome investigations were based on either adult^{7–9,13,14,29–32,38} or paediatric^{10,17} patients, while some studies included only a single sex¹⁴. Geographical region is another crucial factor that has to be considered given that different populations have distinct dietary habits^{39,40}. Moreover, despite the large number of microbiome studies, most did not include external cohorts^{7–10,13,17,29,31,32}. Thus, further studies are necessary to characterize and ascertain microbial alterations in patients with MASLD, MASLD-related cirrhosis or MASLD-related HCC from different populations.

Non-bacterial microbiome in MASLD

Fungi. Commensal fungi are rare members of the gut microbiome, accounting for 0.1–0.5% of the total microorganisms in the human gut^{41,42}. They are often neglected in microbiome studies owing to their low abundance and the lack of well-characterized reference fungal genomes for aligning sequencing reads⁴³. Using fungi-specific internal transcribed spacer sequencing, several studies have profiled the gut fungal community (also known as the mycobiome) in MASLD^{44–47}. Notably, although a range of fungal signatures was identified across studies, consistent enrichment of *Mucor* species (for example, *Mucor ambigua*) was reported in Chinese⁴⁶ and German^{45,47} patients with MASLD. *Mucor* is an opportunistic pathogen that can induce mucormycosis, a serious fungal infection, in immunocompromised patients with uncontrolled type 2 diabetes mellitus or haematological malignancies undergoing chemotherapy⁴⁸. Enriched *Candida albicans*, which is a widely recognized pathogenic fungus⁴⁹, was also observed in patients with MASLD-related fibrosis (stage 2–4 fibrosis)⁴⁵. However,

Table 1 | Human gut microbial signatures in different stages of MASLD progression

Microbial taxa	Simple steatosis	MASH	MASLD-related cirrhosis	MASLD-related HCC
Actinobacteria				
Actinobacteria	Increase ^{14,124}	Decrease ¹⁰	NA	NA
<i>Bifidobacterium</i>	Increase ¹⁴	Decrease ¹⁰	NA	Decrease ²⁹
<i>Eggerthella</i>	Increase ¹⁴	NA	Increase ³²	NA
Bacteroidetes				
Bacteroidetes	Increase ¹⁷ Decrease ^{9,241}	Increase ^{10,17}	NA	NA
<i>Bacteroides</i>	NA	Increase ²⁴²	NA	Increase ²⁹
<i>Prevotella</i>	Increase ^{15,124} Decrease ⁹	Increase ¹⁰ Decrease ²⁴²	NA	NA
Firmicutes				
Firmicutes	Decrease ^{8,14,17}	Decrease ^{10,13}	NA	NA
<i>Acidaminococcus</i>	Increase ¹⁴	NA	Increase ³¹	NA
<i>Blautia</i>	Increase ⁹	Decrease ¹⁰	Increase ³¹ Decrease ³²	Decrease ³²
<i>Clostridium</i>	Increase ^{9,15,17}	NA	NA	NA
<i>Clostridium bolteae</i>	NA	Increase ¹⁶	Increase ^{31,32}	Increase ³²
<i>Coprococcus</i>	Decrease ^{12,14}	Decrease ¹⁰	NA	NA
<i>Coprococcus</i> sp. ART55/1	Increase ³⁸	NA	Decrease ³²	Decrease ³²
<i>Eubacterium</i>	Decrease ¹⁴	Decrease ^{10,13}	NA	NA
<i>Eubacterium eligens</i>	Decrease ¹⁴	NA	Decrease ³¹	NA
<i>Eubacterium rectale</i>	Decrease ¹⁴	NA	Decrease ³¹	NA
<i>Faecalibacterium</i>	Decrease ^{8,9,12,30}	NA	NA	NA
<i>Faecalibacterium prausnitzii</i>	NA	Decrease ^{13,16}	Decrease ^{31,32}	Decrease ³²
<i>Lactobacillus</i>	Increase ⁷	Increase ¹⁷	NA	NA
<i>Megasphaera</i>	Increase ³⁰	NA	Increase ³¹	NA
<i>Oscillibacter</i>	Increase ^{14,17} Decrease ⁷⁹	Decrease ^{10,241}	NA	NA
<i>Roseburia</i>	Increase ⁷	Decrease ¹⁰	NA	NA
<i>Ruminococcus</i>	Decrease ¹²	Increase ²⁴² Decrease ¹⁰	NA	NA
<i>Ruminococcus bicirculans</i>	NA	Decrease ¹⁶	Decrease ³²	Decrease ³²
<i>Ruminococcus gnavus</i>	Increase ¹⁴	NA	Increase ^{31,32}	Increase ³²
<i>Ruminococcus obeum</i>	Decrease ¹⁴	Decrease ¹³	NA	NA
<i>Streptococcus</i>	Increase ^{8,30}	NA	Increase ^{29–31}	Increase ²⁹
<i>Veillonella</i>	NA	NA	NA	NA
<i>Veillonella parvula</i>	NA	NA	Increase ³¹	Increase ³²
Fusobacteria				
<i>Fusobacterium</i>	Increase ^{9,17}	NA	NA	NA
Proteobacteria				
Proteobacteria	Increase ^{7,9,14,17}	Increase ^{10,13,17}	NA	Increase ³²
<i>Escherichia</i>	Increase ^{9,14}	Increase ¹⁰	Increase ³¹	NA
<i>Escherichia coli</i>	Increase ¹⁴	Increase ¹⁶	NA	NA
Verrucomicrobia				
<i>Akkermansia</i>	Increase ^{14,17}	NA	Decrease ²⁹	Decrease ²⁹

Only microbial phyla, genera and species that are altered in at least two MASLD stages or studies, with significant differences compared with healthy or precancerous stages (adjusted $P < 0.05$), are included. HCC, hepatocellular carcinoma; MASH, metabolic dysfunction-associated steatohepatitis; MASLD, metabolic dysfunction-associated steatotic liver disease; NA, not applicable.

a subsequent study identified an enrichment of *C. albicans* in patients with alcohol-related liver disease⁴⁷, indicating that *C. albicans* is not a fungal signature specific to MASLD.

Viruses and bacteriophages. The human gut virome is primarily composed of bacteriophages ($\geq 90\%$), with a small fraction of eukaryotic viruses ($\leq 10\%$)⁵⁰. Infection by hepatitis B virus and hepatitis C virus are well-recognized causes of liver disease, and current evidence has suggested the involvement of gut bacteriophages in MASLD. In a study by Lang et al., a high abundance of *Lactobacillus* phages was identified in German patients with MASLD, and this increase in abundance was positively correlated with severity of fibrosis⁵¹. Depleted *Leuconostoc* and *Lactococcus* phages were also observed⁵¹. Interestingly, although the fundamental role of bacteriophages is to eliminate bacteria, studies have shown an insignificant correlation between viral diversity and bacterial diversity in the gut microbiome⁵¹. Given the emerging therapeutic potential of bacteriophages⁵², it is worth evaluating their effects on gut bacteria during disease development. Nevertheless, virome research is challenged by the individual-specific nature of the viral community as multiple studies have reported that only a maximum of 20% of gut viral species are shared among individuals^{51,53,54}. The great diversity of virome composition, together with the low viral load in the human gut, contamination by bacterial DNA, amplification bias in sequencing and insufficiency of viral sequence databases, are critical issues of current virome investigation^{55,56}.

Intrahepatic or intratumoural microbiome

The human liver has long been regarded as a sterile organ but increasing evidence has reported the presence of an intrahepatic microbial community with functions in physiological and disease conditions. For example, an enrichment of the phylum Proteobacteria was observed in human and mouse liver, and treating mice with broad-spectrum oral antibiotics led to a 90% reduction in hepatic immune cells and to suppressed immune cell recruitment and maturation⁵⁷. Through 16S rRNA gene sequencing, Sookoian et al. identified enriched *Peptostreptococcus*, *Bifidobacterium* and *Roseibacillus* in the liver of patients with MASLD with morbid obesity (body mass index (BMI) $>40 \text{ kg/m}^2$)⁵⁸. By contrast, the intrahepatic microbiome in patients with MASLD with non-morbid obesity is dominated by Proteobacteria, the abundance of which is histologically correlated with ballooning degeneration, lobular and portal inflammation, and liver fibrosis⁵⁸.

In human HCC tumours with mixed aetiologies (including MASLD), a diversity of microbial taxa are enriched such as *Stenotrophomonas*, *Phyllobacterium* and *Enterococcus*⁵⁹ as well as the phylum Actinobacteria⁶⁰. In 2025, Wang et al. confirmed the presence of live *K. pneumoniae* in liver tissues of patients with HCC (mixed aetiologies) and demonstrated that intestinal *K. pneumoniae* could translocate into mouse liver to promote inflammation and tumourigenesis⁶¹. Mechanistically, penicillin-binding protein 1B, the surface protein of *K. pneumoniae*, binds to TLR4 on HCC tumour cells, causing increased cell proliferation and activation of oncogenic signalling pathways. Other clinical studies have illustrated the potential of intratumoural microorganisms as diagnostic⁶² or prognostic^{63,64} biomarkers of HCC caused by viral hepatitis and other non-MASLD aetiologies. By comparison, the intratumoural microbiome specific to MASLD-related HCC is less characterized. A human study reported the significant enrichment of intrahepatic bacteria, particularly of the families *Ruminococcaceae* and *Bacteroidaceae*, in MASLD-related HCC and MASLD-related cirrhosis, compared with patients with non-cirrhotic

MASLD⁶⁵. However, it is worth noting that current investigations on the intrahepatic microbiome are mostly sequencing studies. Although many studies have revealed enriched intrahepatic bacterial DNA fragments in disease conditions, only a few provide direct evidence of the presence of live bacteria in human liver by tissue culture^{59,61,62}. Moreover, the mechanistic function of intratumoural microorganisms or their DNA fragments in MASLD-related HCC remains completely unknown.

Risk factors of gut microbial dysbiosis in MASLD

Certain diets are associated with an increased risk of MASLD. An example is the Western diet, which involves high contents of saturated fats and refined carbohydrates^{66,67}. Excess energy from carbohydrates and fats is converted to triglycerides in the liver, whereas a net accumulation of triglycerides subsequently promotes MASLD pathogenesis by inducing hepatic injury, inflammation and steatosis⁶⁸. By contrast, the Mediterranean diet, which is rich in fibres, fruits and whole grains, is recommended for the management of MASLD by European guidelines⁶⁹ as it can lower the risk of MASLD by improving insulin sensitivity^{66,70–72}. Unhealthy diets also result in gut microbial dysbiosis. A murine study reported that excess consumption of fats and cholesterol leads to spontaneous and sequential progression from liver steatosis, to fibrosis and to HCC in a microbiome-dependent manner⁷³. By contrast, MASH progression in mice could be alleviated by consumption of the soluble fibre inulin, which is absorbed by the gut commensal *Parabacteroides distasonis* to produce the anti-steatotic metabolite pentadecanoic acid⁷⁴. In addition to diet, individuals with unhealthy lifestyles, lack of physical activity, alcohol use disorder and who smoke are also predisposed to an increased risk of MASLD⁷⁵. Intriguingly, Chen et al. illustrated that *Bacteroides xylanisolvens*, a species that is depleted in patients with MASLD, was able to degrade gut nicotine to protect mice from smoking-induced MASLD progression⁷⁶.

Mechanisms of dysregulated gut–liver axis in MASLD and MASLD-related HCC

The pathogenesis of MASLD is an extremely complicated, multifactorial process. The underlying mechanisms involve triglyceride accumulation and lipotoxicity, insulin resistance, and dysregulated nuclear receptor and immune signalling^{68,77}. Emerging studies have also demonstrated how the crosstalk between host genetics or epigenetics and the gut microbiome contributes to MASLD progression, highlighting the role of host–microorganism interactions in modulating lipid metabolism, inflammation and fibrogenesis^{78,79}. In this section, we explore the major pathogenic mechanisms that are related to the gut microbiome in MASLD, including intestinal barrier dysfunction, chronic inflammation, and dysregulated metabolic pathways and metabolites (Fig. 2).

Impaired intestinal barrier

The gatekeeping function of the intestinal barrier is greatly disrupted in liver diseases. Elevated blood levels of endotoxins, especially lipopolysaccharides (LPS; the main component of the outermost membrane in Gram-negative bacteria)^{80–82}, as well as LPS-binding protein (LBP)⁸³, are hallmark features of MASLD and MASH. Increased intestinal permeability concomitant with decreased expression of tight junction proteins (such as claudins, junctional adhesion molecule A, occludin and zonula occludens 1), were also observed in patients with MASLD^{82,84,85} and mice with diet-induced MASH⁸⁶. In particular, Mouries et al. observed that 1-week feeding of a high-fat diet (HFD; a classic mouse model of diet-induced MASLD) is sufficient to increase intestinal permeability in mice⁸⁷. Through faecal microbiota transplantation (FMT), increased

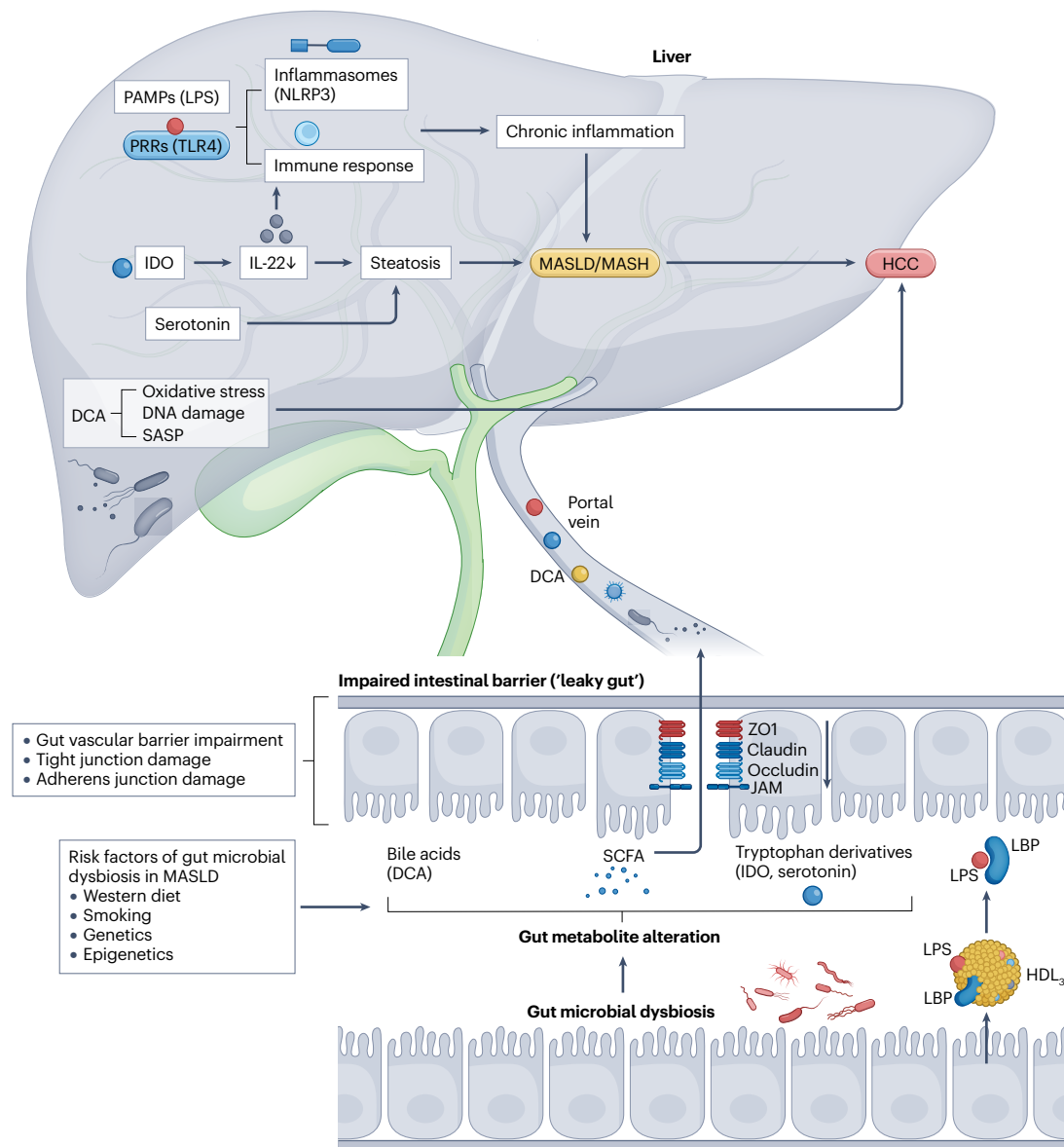


Fig. 2 | Gut microbiome-mediated mechanisms in MASLD. Risk factors such as a Western diet, unhealthy lifestyle (for example, smoking), and genetic and epigenetic factors promote metabolic dysfunction-associated steatotic liver disease (MASLD) by dysregulation of the gut–liver axis via various mechanisms. First, an impaired intestinal barrier. The gut microbiome and metabolites can compromise the intestinal barrier, leading to increased intestinal permeability with reduced expression of junctional proteins (for example, claudins, junctional adhesion molecule A (JAM), occludin and zonula occludens 1 (ZO1)). Second, dysregulated metabolic pathways and metabolites. Altered gut microbial-derived metabolites, including bile acids, short-chain fatty acids (SCFAs) and tryptophan derivatives, contribute to the onset and progression of MASLD. The secondary bile acid deoxycholic acid (DCA) can induce DNA damage, mitochondrial oxidative stress and a senescence-associated secretory phenotype (SASP) in hepatic stellate cells to accelerate MASLD-related hepatocellular carcinoma (HCC) development. Indoleamine 2,3-dioxygenase

(IDO) is the rate-limiting enzyme of kynurenine biosynthesis that promotes liver steatosis and inflammation by inhibiting IL-22. Serotonin is another product of tryptophan metabolism that also promotes liver steatosis. Third, chronic inflammation. An impaired intestinal barrier enables increased translocation of pathogen-associated molecular patterns (PAMPs; such as lipopolysaccharide (LPS)) into the bloodstream. After reaching the liver, PAMPs interact with pattern recognition receptors (PRRs; such as Toll-like receptor 4 (TLR4)) to trigger an inflammatory response, including inflammasome activation (mainly NOD-, LRR- and pyrin domain-containing 3 (NLRP3)) and innate immunity. In normal conditions, high-density lipoprotein subspecies 3 (HDL₃) derived from enterocytes blocks the binding of LPS to LPS-binding protein (LBP), whereas dysregulated HDL₃ in MASLD loses its blocking capacity, enabling LPS–LBP binding to promote liver inflammation. MASH, metabolic dysfunction-associated steatohepatitis.

intestinal permeability was observed only in germ-free mice transplanted with stools from HFD-fed mice but not in those receiving stool from mice fed a standard diet, suggesting that HFD-induced microbial dysbiosis is the cause rather than the consequence of intestinal barrier impairment⁸⁷.

Notably, the underlying mechanisms of how an altered microbiome leads to increased intestinal permeability and intestinal barrier dysfunction are largely unexplored. It has been speculated that the intestinal barrier is readily influenced by metabolites derived from gut microorganisms as several beneficial metabolites, such as SCFAs, indole derivatives and urolithin A, are capable of improving intestinal barrier integrity^{88–90}. In 2023, one study identified that ethanolamine, an essential phospholipid constituent of the human cell membrane and energy source for several gut bacteria (for example, *Clostridium*, *Enterococcus* and *Escherichia*)^{91,92}, is accumulated in the gut of obese mice, together with depleted levels of ethanolamine-metabolizing bacteria⁹³. In turn, elevated ethanolamine upregulates the expression of microRNA-101a-3p to reduce the mRNA stability of tight junction proteins, thereby suppressing protein translation and weakening the intestinal barrier. Nonetheless, metabolites or other microbiome-associated mechanisms that promote intestinal barrier dysfunction in MASLD pathogenesis remain undetermined.

Dysregulated gut microbiome-mediated metabolic pathways and metabolites

Metabolites produced by gut commensal microorganisms are also critically involved in the pathogenesis and progression of MASLD. Among the large array of metabolites, the most studied categories of microbial-mediated metabolites in MASLD include bile acids, SCFAs, tryptophan derivatives and endogenous ethanol.

Bile acid metabolism. Elevated levels of DCA and its derivatives, including ursodeoxycholic acid (UDCA) and glycodeoxycholic acid, were consistently observed in serum and stool of patients with MASLD and MASH in different studies^{12,94,95}, implicating secondary bile acids as key modulators of gut–liver axis dysregulation. Serum, faecal and intrahepatic DCA levels are also significantly increased in mice with MASLD-related HCC induced by carcinogens and a HFD^{96,97}. DCA is a cytotoxic bile acid primarily produced by gut commensal *Clostridium* clusters XI and XIVa via 7 α -dehydroxylation⁹⁸. In mice with carcinogen-induced and HFD-induced MASLD-related HCC, increased DCA levels are positively correlated with *Clostridium* abundance, whereas eradicating these bacteria with narrow-spectrum antibiotics reverses DCA elevation and dampens tumour development⁹⁶. Mechanistically, intrahepatic DCA provokes the senescence-associated secretory phenotype in hepatic stellate cells (HSCs), causing them to secrete more pro-tumorigenic factors and prostaglandin E₂ (PGE₂) to suppress antitumour immunity and exacerbate MASLD-related HCC development in mice^{96,99}. Elevated DCA also induces cytotoxicity to intestinal epithelial cells, which in turn destabilizes the intestinal barrier and enables more gut pathobionts and their derivatives to reach the mouse liver^{100,101}. Interestingly, a human study reported that both primary and secondary bile acids are increased in the serum of patients with MASH⁹⁴. The elevation of primary bile acids is due to an almost tenfold upregulation of rate-limiting enzymes for bile acid synthesis (for example, CYP7A1) in the liver. Meanwhile, there were no significant changes in the abundance of bile acid hydrolase-expressing bacteria (for example, *Clostridium* and *Bacteroides*), which are responsible for converting primary to secondary bile acids. Instead, the increase in

secondary bile acids in patients with MASH is attributed to the enrichment of gut bacteria that directly metabolize taurine and glycine in bile acids, particularly *Escherichia* and *Bifidobacteria*.

Certain gut microorganism-mediated bile acids inhibit hepatocarcinogenesis. Using multiple mouse models of non-MASLD-related HCC, Ma et al. identified that natural killer T (NKT) cells inhibit tumour growth by rapid secretion of interferon- γ (IFN γ), while hepatic accumulation of NKT cells is mediated by CXC-chemokine ligand 16 (CXCL16)¹⁰². The expression of CXCL16 is regulated by bile acids: the primary bile acid chenodeoxycholic acid can upregulate CXCL16 to enhance hepatic accumulation of NKT cells, whereas secondary bile acids (for example, taurothiocholic acid) have opposite effects. Notably, the antitumour function of NKT cells activated by primary bile acids and CXCL16 is dependent on the gut microbiome. Blocking bacterial conversion of primary to secondary bile acids with antibiotics could increase NKT cell accumulation and decrease tumour growth in mouse liver¹⁰². By contrast, direct colonization of 7 α -dehydroxylation-mediating *Clostridium* species promotes tumour growth, with reduced hepatic NKT cells in mice¹⁰².

Owing to the advances in metabolomic profiling, an increasing number of bile acids, especially those synthesized under unconventional pathways, have been implicated in MASLD progression. For example, Kuang et al. reported reduced serum levels of hyodeoxycholic acid (HDCA), a non-12 α -hydroxylated bile acid, in patients with MASLD and in mice with chemical-induced and/or HFD-induced MASLD¹⁰³. HDCA inhibits liver steatosis in mice by suppressing intestinal farnesoid X receptor (FXR) signalling and stimulating the gut commensal *P. distasonis* to generate more γ -linolenic acid, resulting in a shift from the classical CYP7A1 to the alternative CYP7B1 pathway of bile acid synthesis. In 2024, Nie et al. used a multistep strategy to identify microorganism-derived bile acids by incubating a synthetic alkyne probe of cholic acid with microbial communities isolated from human stools, followed by mass spectrometry detection¹⁰⁴. The result reveals a reduction of previously uncharacterized 3-succinylated cholic acid in patients with MASLD, which is produced by β -lactamase-expressing *Bacteroides uniformis*. This secondary bile acid mitigates the development of diet-induced MASLD in mice with an intact gut microbiome by promoting growth of the beneficial gut commensal *Akkermansia muciniphila* and restoring intestinal barrier integrity.

Short-chain fatty acids. Dietary fermentation of indigestible carbohydrates is mediated by gut bacteria (for example, *Bacteroides*, *Bifidobacterium*, *Clostridium*, *Eubacterium* and *Faecalibacterium*), which generates SCFAs, including acetate, propionate and butyrate, as end products. SCFAs are pivotally associated with liver function and metabolism, contributing to energy expenditure and lipid oxidation¹⁰⁵. SCFAs are generally protective against MASLD as evidenced by a 24-week administration of propionate resulting in reduced hepatic lipid accumulation in adults with overweight¹⁰⁶. In preclinical studies, feeding mice with the soluble fibre inulin leads to an enrichment of SCFA-producing bacteria together with elevated portal vein levels of acetate¹⁰⁷ and propionate¹⁰⁸. Increased acetate then binds to hepatocyte-free fatty acid receptor 2 (FFAR2)¹⁰⁷ or propionate activates the AMP-activated protein kinase (AMPK) signalling pathway¹⁰⁸, both thereby suppressing MASLD development in mice. Similarly, butyrate also activates the anti-steatotic AMPK pathway to attenuate liver steatosis, reduce triglyceride accumulation and improve insulin sensitivity in mice with HFD-induced MASLD^{109,110}. At the epigenetic level, both acetate and butyrate inhibit histone deacetylase (a major mediator

of histone modification, the aberrant activation of which is associated with many liver diseases¹¹¹ to alleviate hepatic lipid lipotoxicity and inflammation in diabetic rats¹¹² as well as to induce cell death of pro-inflammatory macrophages in mice with diet-induced MASH¹¹³. In addition, probiotics, such as *Bifidobacterium pseudolongum* and *Lactobacillus acidophilus*, produce SCFAs, thereby downregulating oncogenic signalling pathways and suppressing tumorigenesis in mice with carcinogen-induced and diet-induced MASLD-HCC^{114,115}.

By contrast, SCFAs can promote disease development if doses exceed the threshold tolerable by the host^{116,117}. Singh et al. identified that a mixture of SCFAs at high concentrations aggravates hepatic lipogenesis and insulin resistance in TLR5-deficient mice¹¹⁸. A subsequent study showed that supplementing high-concentration butyrate (100 mM) to TLR5-deficient mice causes liver inflammation and creates a microenvironment that favours HCC development¹¹⁹. On the other hand, ablation of G protein-coupled receptor 41 (GPR41) and GPR43, the two dominant hepatocytic SCFA receptors, improves energy expenditure and reduces adiposity and hepatic triglycerides in HFD-fed mice^{120,121}. Owing to the varying effects of SCFAs, it has been speculated that their function in MASLD might be dependent on the pathway or mechanism that they activate such as AMPK, G protein-coupled receptors and histone deacetylase activity¹²².

Similarly to the inconsistent mechanistic findings, SCFA concentrations in MASLD have varied across human studies. Although some studies reported increased faecal SCFAs in patients with MASLD^{13,123}, no significant changes were observed in other studies^{12,124}. Moreover, faecal metabolites might not actually represent metabolomic alterations in the human body as exemplified by a study reporting that SCFA concentrations only increase in stool but not in the serum of patients with MASLD¹³. Another study using paired faecal and serum samples identified that faecal and blood metabolomes are independent of each other in MASLD and other cardiometabolic diseases, highlighting sample type as a determining factor when interpreting metabolomic data¹²⁵. In addition, Behary et al. revealed the marked enrichment of faecal and serum SCFAs and SCFA-producing bacteria in patients with MASLD-related HCC from a single cohort without external validation³². To confirm these observations, future studies should consider metabolomic changes in paired serum and faecal samples from multiple cohorts as well as microbial alterations to validate the correlation between SCFAs and SCFA-producing bacteria in MASLD.

Tryptophan metabolism. Tryptophan is an essential amino acid necessary for protein biosynthesis and metabolism. Over 90% of dietary tryptophan follows the kynurenine pathway, whereas the remaining undergoes the serotonin (also known as 5-hydroxytryptamine) or indole pathway^{126,127}. In MASLD, these tryptophan metabolism pathways are dysregulated as evidenced by elevated serum and faecal levels of kynurenine^{128,129} and serotonin^{130,131} in patients. Kynurenine is notorious for its contribution to tumour progression and immune escape¹³², whilst indoleamine 2,3-dioxygenase, the rate-limiting enzyme of kynurenine biosynthesis, can be activated by LPS and pro-inflammatory cytokines¹³³. Ablation of indoleamine 2,3-dioxygenase is associated with decreased liver steatosis, inflammation and endotoxaemia in HFD-fed mice through microbiome-dependent production of IL-22 as well as improved intestinal barrier integrity and insulin sensitivity¹³⁴. Serotonin production is mediated by tryptophan hydroxylase and 5-hydroxytryptamine receptors, whereas knockout or pharmacological inhibition of these regulators ameliorated HFD-induced liver steatosis in mice, accompanied by reduced liver inflammation and lipid

accumulation^{130,135}. In the context of energy expenditure, serotonin is capable of blunting β -adrenergic induction of the thermogenic programme in brown and beige adipocytes, thereby accelerating MASLD development in HFD-fed mice¹³⁶.

Unlike the kynurenine and serotonin pathways, which are regulated by both host cells (for example, immune cells and intestinal enterochromaffin cells, respectively) and the gut microbiome, indole metabolism is primarily dependent on gut bacteria that express the enzyme tryptophanase, including *Clostridium*, *Bacteroides*, *Escherichia*, *Bifidobacterium* and *Lactobacillus*^{127,137}. These bacteria metabolize dietary tryptophan into various indole derivatives such as indole-3-aldehyde, indole-3-acetic acid (IAA), indole-3-propanoic acid and indole-3-carboxylic acid. Many indole derivatives are ligands for the aryl hydrocarbon receptor, a signalling pathway pivotal for maintaining homeostasis of the intestinal barrier and immune cells^{138,139}. Meanwhile, clinical evidence has indicated a marked depletion of serum and faecal indole derivatives in patients with MASLD^{129,140,141} as well as in mice with diet-induced MASLD-related HCC⁷³. Supplementation with indole, IAA or indole-3-propanoic acid significantly alleviated the severity of diet-induced MASLD in mice by attenuating liver steatosis, inflammation, fibrosis, lipogenesis and endotoxaemia^{140,142–144}. Moreover, decreased levels of intestinal IAA and increased levels of kynurenine were observed in patients with obesity or type 2 diabetes mellitus and in HFD-fed mice, implying a shift in tryptophan metabolism from the beneficial indole pathway towards production of harmful kynurenine under conditions of metabolic disease¹³⁴.

Endogenous ethanol. Endogenous ethanol is produced mainly by yeast (particularly *Saccharomyces cerevisiae*) and certain bacteria (for example, *E. coli*, *K. pneumoniae* and *Staphylococcus aureus*) in the human gut microbiome¹⁴⁵. In Chinese patients with simple steatosis, blood alcohol concentration (BAC) was markedly increased to 224 mg/dl or even higher in MASH (400 mg/dl) compared with non-MASLD controls (62 mg/dl)¹⁴⁶. Portal vein ethanol concentration was also higher in European patients with simple steatosis (8.0 mM) or MASH (21.0 mM) than in non-MASLD controls (2.1 mM), and an intervention study showed that the increase in portal and peripheral BAC is associated with dietary stimuli¹⁴⁷. These findings indicate that, in MASLD, the liver is chronically exposed to excess amounts of ethanol even without alcohol consumption, suggesting a pathogenic role for microorganism-derived endogenous ethanol. Through metagenomic sequencing, Yuan et al. identified a positive correlation between BAC and faecal abundance of *K. pneumoniae* in patients with MASLD, while supplementation of ethanol-producing *K. pneumoniae* accelerates MASLD by inducing severe liver injury, inflammation, fibrosis and mitochondrial dysfunction in mice¹⁴⁶. By contrast, a prospective study revealed that lactic acid-producing *Lactobacillus* species, rather than *K. pneumoniae*, are positively correlated with BAC in patients with MASLD¹⁴⁷. Such a disparity might be explained by the differences in ethnicity and geographic region between these two studies. In addition to gut bacteria, certain fungal *Candida* species (for example, *C. albicans* and *Candida glabrata*) also produce excess ethanol in patients with MASH¹⁴⁸.

Chronic inflammation induced by PAMPs

PAMPs are structural molecules derived from gut pathobionts, and they can activate innate pattern recognition receptors (PRRs) to initiate a downstream pro-inflammatory cascade. The most studied PAMP is LPS, while others include nucleic acids, and surface proteins and

membrane components of gut bacteria and viruses. In patients with MASLD, blood LPS level is elevated^{80–82}, while increased exposure to LPS in the liver can lead to local inflammation. Chronic inflammation is fundamental to MASLD pathogenesis. Numerous pro-inflammatory events, including inflammasome activation and a dysregulated innate immune response, together form an interdependent network to induce persistent liver inflammation and accelerate MASLD progression⁶⁸. Via the impaired intestinal barrier, more PAMPs can reach the liver and activate PRRs to induce local inflammation. Of the different PRRs, TLRs, especially TLR4, are the predominant contributors to PAMP-induced inflammation. An early animal study reported that liver inflammation and fibrogenesis are markedly alleviated in mice deficient in TLR4 or its downstream effector myeloid differentiation primary response protein 88 (MyD88)¹⁴⁹. TLR4 in quiescent HSCs was identified as the main target of LPS, and LPS-induced TLR4 upregulation promotes unrestricted activation of Kupffer cells, with constant secretion of pro-inflammatory chemokines¹⁴⁹. Moreover, LPS-induced TLR4 activation in Kupffer cells¹⁵⁰ and hepatocytes¹⁵¹ can also exacerbate liver inflammation in mice.

The NOD-, LRR- and pyrin domain-containing 3 (NLRP3) inflammasome has been implicated in MASLD progression. In response to PAMPs, the NLRP3 inflammasome is formed to activate caspase 1, which then induces the release of pro-inflammatory IL-1 β and IL-18 (ref. 152). In MASH, the expression of inflammasome genes is significantly upregulated in patients¹⁵³, whereas preclinical evidence illustrated that ablation of NLRP3 protects mice from diet-induced steatohepatitis, with reduced liver injury and infiltration of activated macrophages^{154,155}. Notably, the function of the NLRP3 inflammasome can be altered by the gut microbiome as several studies reported that NLRP3 inflammasome deficiency leads to worsened liver steatosis, inflammation and injury in mice with diet-induced MASH in a microbiome-dependent manner^{156,157}. For example, after co-housing for 4 weeks, wild-type mice develop comparable levels of liver injury and inflammation to co-housed inflammasome-deficient mice¹⁵⁶. This increase in MASH severity is attributed to the elevated efflux of bacteria-derived TLR4 and TLR9 agonists (mainly from *Porphyromonadaceae*) from mouse intestines into the liver, where they activate TLR4 and TLR9, respectively, to exacerbate liver inflammation and MASH progression. In addition to the gut microbiome, the controversial role of the NLRP3 inflammasome can also be explained by the disparity among cell types as diet-induced liver inflammation and steatosis are alleviated only in mice with myeloid-specific *Nlrp3* ablation but not in mice with hepatocyte-specific or HSC-specific *Nlrp3* knockout¹⁵⁸.

In normal conditions, PAMPs are counteracted and cleared by Kupffer cell-mediated phagocytosis, hepatocyte-mediated biliary excretion¹⁵⁹ and binding to high-density lipoprotein (HDL)¹⁶⁰. For example, HDL subspecies 3 (HDL₃) derived from enterocytes can block the binding of LPS to PRRs on macrophages in the portal vein, protecting mice from liver inflammation and fibrosis induced by surgery, diet or alcohol consumption¹⁶¹. By interacting with LBP, the HDL₃–LBP complex inactivates and masks LPS from detection by PRRs, and is then removed from the liver. Notably, Han et al. reported that only HDL₃ exhibits robust LPS-neutralizing function in mice¹⁶¹ but clinical evidence showed that MASLD severity is negatively correlated with serum levels of other HDL subspecies but not HDL₃ (ref. 162). Moreover, the reasons why PAMP-neutralizing mechanisms become less effective or even dysfunctional in MASLD and MASH are unknown. It is possible that these protective mechanisms are dysregulated by the dysbiotic gut microbiome in these diseases and/or by being overwhelmed by the

amount of PAMPs in the portal circulation due to increased intestinal permeability.

LPS and TLR4 signalling continue to have crucial pathogenic roles during the progression to liver malignancy. In patients with HCC with mixed aetiologies (including MASLD), higher expression of TLR4 in tumour tissues is significantly more associated with early recurrence and poor survival compared with patients with lower TLR4 expression^{163,164}. LPS-induced TLR4 activation contributes to hepatocarcinogenesis by promoting epithelial–mesenchymal transition, metastasis and growth of tumour cells in patients with HCC^{163–165}. LPS–TLR4 signalling also induces the differentiation of hepatic progenitor cells into myofibroblasts, with elevated secretion of pro-inflammatory IL-6 and tumour necrosis factor (TNF), thereby creating a microenvironment that favours malignant transformation of progenitor cells into tumour cells in mice¹⁶⁶. In 2023, a study identified that *Stenotrophomonas maltophilia*, a multidrug-resistant bacterium that can cause a wide range of infections in patients with cancer^{167,168}, is enriched in human HCC tumours⁵⁹. Once colonized in the liver, *S. maltophilia* activates the TLR4–nuclear factor- κ B (NF- κ B) signalling pathway in senescent HSCs, leading to formation of the NLRP3 inflammasome complex, with increased secretion of pro-inflammatory cytokines (for example, IL-1 β and IL-6), and eventually accelerating HCC progression in mice. On the other hand, the pro-tumorigenic function of *S. maltophilia* and TLR4 was markedly abolished by the use of NLRP3 inhibitor⁵⁹. Owing to the significance of TLR4, it is worth investigating the therapeutic potential of TLR4 inhibitors to treat MASLD and its related diseases.

Effects of the gut microbiome and metabolites on immune responses in MASLD and MASLD-related HCC

In addition to PAMPs, the gut microbiome itself can promote MASLD progression by direct modulation of host immunity. For example, increased levels of activated B cells were present in the liver of mice with HFD-induced MASLD, and these B cells induced innate MyD88 signalling to promote intrahepatic infiltration of IFN γ ⁺ T cells and secretion of pro-inflammatory IL-6 and TNF¹⁶⁹. Notably, intrahepatic accumulation and activation of B cells were recapitulated by transplanting stools from patients with MASLD into the recipient microbiome-depleted mice, implying that gut microorganisms drive the pro-inflammatory function of B cells to stimulate the innate response and contribute to MASH. By contrast, a later study reported that the effect of B cell activation on MASH is in fact associated with IgA rather than with the gut microbiome¹⁷⁰. In particular, HFD-fed germ-free mice still developed full-blown MASH, inflammation and impaired glucose response, whereas MASH in these germ-free mice was rescued by therapeutic depletion of B cells, indicating that the gut microbiome is not necessary for diet-induced MASH development¹⁷⁰. In addition to B cells, Miyamoto et al. identified a subset of immunosuppressive macrophages that are specifically enriched in the portal circulation and these IL-10-expressing and MARCO-expressing macrophages are significantly depleted in patients with MASH, compared with healthy individuals without MASH¹⁷¹. Activation of these macrophages in the portal vein is dependent on gut microbial metabolites, particularly isoallothiocholic acid derived from *Odoribacteraceae*, whereas the deficiency of these periportal macrophages causes intestinal barrier leakage and further exaggerates liver inflammation in mice¹⁷¹.

An increasing number of studies have focused on immune populations in MASLD-related HCC. Unlike other HCC tumours, the tumour microenvironment of MASLD-related HCC has an expanded population of exhausted programmed cell death 1 (PD1)⁺CD8⁺ T cells, which

impaired immune surveillance and reduced survival in patients receiving anti-PD1 immunotherapy¹⁷². A subsequent single-cell study also reported increased intratumoural infiltration of exhausted PD1⁺CD8⁺ T cells, myeloid-derived suppressor cells and tumour-associated macrophages in patients with MASLD-related HCC but not in human HCC tumours induced by viral hepatitis¹⁷³. Through culturing peripheral blood mononuclear cells with faecal samples from patients, Behary et al. revealed that the gut microbiome of MASLD-related HCC is enriched with butyrate-producing bacteria and positively associated with an immunosuppressive response³². In particular, faecal extracts from patients with MASLD-related HCC promote the expansion of immunosuppressive IL-10⁺ regulatory T cells in vitro, while attenuating anti-tumour CD8⁺ T cells, monocytes and B cells. Notably, while current evidence has suggested a correlation between the gut microbiome and alterations in the tumour immune microenvironment, it remains unclear how gut microorganisms suppress antitumour immunity to promote MASLD-related HCC development. It is possible that intratumoural microorganisms might restrict immune cell recruitment and maturation to favour an immunosuppressive tumour microenvironment as demonstrated in a preclinical mouse study of non-MASLD-related HCC⁵⁷.

Clinical applications of the gut microbiome in MASLD and MASLD-related HCC

Diagnosis of MASLD by gut microbial and metabolite biomarkers

Accurate identification of MASH and advanced MASLD-related fibrosis is crucial for stratifying patients who require pharmacological therapies¹⁷⁴. Although liver biopsy remains the key reference for diagnosis, its invasiveness, poor acceptability, high cost and inter-observer variability have limited its routine clinical use¹⁷⁴. Hence, there is a pressing need for noninvasive diagnostic biomarkers with high sensitivity and specificity. Current research has highlighted the potential of microbiome-based diagnostic biomarkers for MASLD. Using metagenomic sequencing, a prospective single-cohort study of 86 patients from the USA with biopsy-proven MASLD found that a set of 40 features, including 37 faecal bacterial species, could discriminate advanced MASLD-related fibrosis from MASH with high accuracy (area under the ROC curve (AUC) 0.936)¹³. A subsequent prospective study further identified a panel of 30 features, including 27 faecal bacterial biomarkers, to detect MASLD-related cirrhosis (AUC 0.87–0.92) in 203 patients from the USA with well-characterized MASLD from two independent cohorts³⁰. Another prospective study reported a panel of 19 faecal bacterial biomarkers that, when combined with age and serum levels of albumin or aspartate aminotransferase (AST), achieves an AUC of 0.91–0.95 to detect MASLD-related cirrhosis in both American and Chinese populations³¹. In addition to gut microorganisms, a panel of 17 faecal metabolites also demonstrated high accuracy (AUC 0.91) to distinguish patients with MASLD-related cirrhosis from non-MASLD controls³¹.

Beyond bacterial biomarkers, the diagnostic potential of gut viruses and fungi in MASLD has also been explored. A model incorporating viral diversity index along with age and serum AST levels could identify severe MASH (MAS score 5–8) or MASLD-related cirrhosis (AUC 0.95) and stage 2–4 MASLD-related fibrosis (AUC 0.88) from a prospective single cohort of 73 German patients with biopsy-proven MASLD⁵¹. Faecal fungal signatures consisting of *Scopulariopsis*, *Kluyveromyces*, *Malassezia restricta* and *Mucor* also showed high accuracy (AUC 0.93) in differentiating MASLD from alcohol-related liver disease in a prospective single-cohort German study⁴⁷.

Noninvasive approaches for diagnosing MASLD mainly include biomarker detection (for example, ELF panel¹⁷⁵, NIS4 panel¹⁷⁶ and SomaSignal¹⁷⁷) and imaging (for example, transient elastography and magnetic resonance elastography). Imaging is advantageous for having high diagnostic accuracy, yet these techniques are operator-dependent and expensive, thus limiting their clinical application¹⁷⁴. Meanwhile, biomarker detection is more convenient and less costly, with easy interpretation of the results. It is noteworthy that, currently, biomarkers for MASLD diagnosis are mostly blood based, whereas the application of microbial biomarkers is still under investigation. Although certain microbial panels did show high accuracy, only a few of them were externally validated^{30,31}. Given that the gut microbiome is readily influenced by diet and geographic region, it remains questionable whether the diagnostic performance of microbial biomarkers can be reproducible in different populations. Moreover, the detection of gut microorganisms and metabolites requires advanced technology such as metagenomics and metabolomics, which are sophisticated methodologies and not suitable for routine clinical practice. By comparison, the detection of serum biomarkers is much more robust and user-friendly. For example, a prospective multi-population study in 2025 reported that high accuracy for MASLD diagnosis could be achieved by detecting two serum biomarkers using a basic assay (that is, enzyme-linked immunosorbent assay) concomitant with BMI¹⁷⁸. Hence, microbial biomarkers for diagnosing MASLD still require extensive optimization before being implemented into clinical practice.

Gut microorganisms can serve as biomarkers for evaluating treatment response in patients with HCC. A prospective multi-cohort study involving 94 Chinese patients with HCC identified that patients with higher faecal abundance of *Veillonella*, *Lachnoclostridium* and the family *Lachnospiraceae* at baseline are more responsive to anti-PD1 immunotherapy (nivolumab and pembrolizumab)¹⁷⁹. In particular, the coexistence of *Lachnoclostridium* enrichment and *Prevotella* depletion is a significant predictor of better overall survival in these anti-PD1-treated patients with HCC. Another prospective single-cohort trial with a small sample size (11 Italian patients with HCC) reported that higher faecal abundance of *Akkermansia* at baseline is associated with better response to anti-PD1 immunotherapy (tremelimumab and/or durvalumab)¹⁸⁰. This finding is consistent with other studies reporting the potential of *Akkermansia* to predict anti-PD1 treatment response in patients with other cancers¹⁸¹. In addition to immunotherapy, gut bacteria, such as *Blautia*, can also serve as predictive biomarkers for chemotherapy response; higher faecal abundance of *Blautia* was associated with better prognosis and survival in 32 Japanese patients with chemotherapy-treated HCC from a prospective multi-cohort study¹⁸². Nonetheless, the performance of microbial biomarkers for predicting treatment response in patients with MASLD-related HCC remains completely undetermined.

Therapeutic modulation of the gut microbiome to treat MASLD

Emerging evidence has shown promise in translating microbiome research into novel therapies for MASLD. Microbiome-targeting approaches, including FMT, probiotics and engineered bacteria, prebiotics and postbiotics, antimicrobials, microbial-derived metabolites and bacteriophages (Fig. 3), have shown encouraging outcomes in preclinical studies and small clinical trials (Table 2).

Faecal microbiota transplantation. FMT involves the transfer of faecal matter from healthy donors into recipients to restore their gut

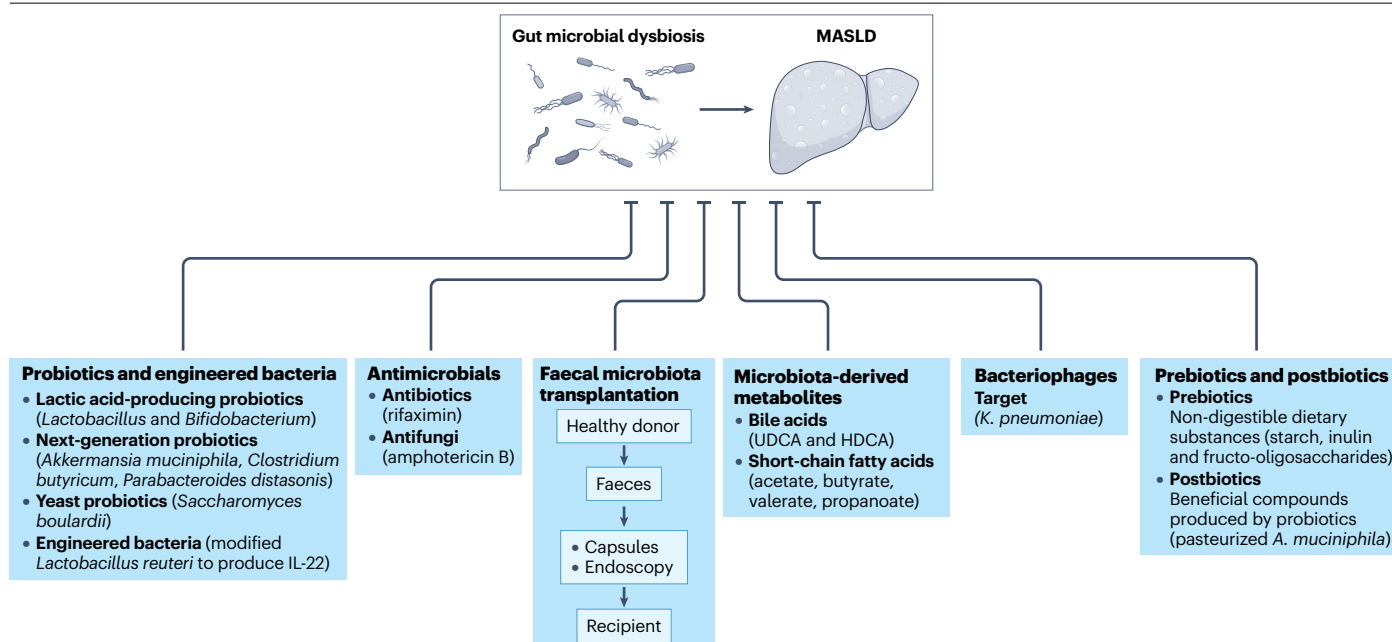


Fig. 3 | Clinical implications of the gut microbiome in MASLD treatment.

Several microbiome-targeting interventions have shown encouraging outcomes to treat and improve metabolic dysfunction-associated steatotic liver disease (MASLD). First, faecal microbiota transplantation, which refers to the transfer of faecal matter from healthy donors to recipients through oral capsules or endoscopy for disease management. Second, probiotics, which are live beneficial bacteria, such as traditional lactic acid probiotics, next-generation strains, yeast probiotics and bioengineering bacteria, that can produce beneficial metabolites.

Third, prebiotics and postbiotics, which are non-digestible food substances that nourish the growth of gut commensal probiotics and their production of beneficial compounds. Fourth, antimicrobials, which directly target specific microorganisms via antibiotics or antifungal treatment. Fifth, microbial-derived metabolites, especially bile acids and short-chain fatty acids, which are critical for human health. Finally, bacteriophages, which target specific bacterial strains (for example, *Klebsiella pneumoniae*-targeting phages) to confer therapeutic benefits. HDCA, hyodeoxycholic acid; UDCA, ursodeoxycholic acid.

microbiome. Preclinical studies have reported the efficacy of FMT to suppress liver inflammation, reduce triglyceride accumulation, promote the growth of gut commensal probiotics and improve intestinal barrier integrity in mice with HFD-induced MASH¹⁸³. However, clinical studies have yielded mixed results. A randomized controlled trial (RCT) enrolling 75 patients with MASLD found that 4-week FMT intervention only reduces liver fat accumulation in patients with MASLD without obesity (253.4 dB/m baseline to 235.7 dB/m after FMT) but not in those with MASLD with obesity (285.2 dB/m baseline to 279.5 dB/m after FMT)¹⁸⁴. In another study involving 21 patients with MASLD, allogenic FMT only reduces small intestinal permeability but does not improve liver steatosis¹⁸⁵. Intriguingly, two clinical trials reported improved liver inflammation and reduced DNA methylation in patients with MASLD with obesity receiving stools from vegan donors without obesity, suggesting that the diet of the donors might be associated with FMT efficacy in the recipients^{186,187}. Moreover, clinical trials have demonstrated that oral FMT capsules are safe and well tolerated in patients with MASLD, end-stage hepatic cirrhosis and encephalopathy¹⁸⁸.

Clinical application of FMT is limited by the lack of a universally defined 'healthy' microbiome as well as by the potential risks associated with the transfer of uncharacterized microbial species (for example, viruses, bacteriophages and fungi)¹⁸⁹. To ensure the safest possible procedure, a panel of the European FMT Working Group recommended a further check of recent history and risk factors of donors on the same day of donation¹⁸⁹. Nevertheless, FMT only shows mild or even no effects in the treatment of MASLD in patients from single-centre studies

with small sample size^{184–187}. These unsatisfying results have raised concerns about the need to conduct larger studies, and there is currently no ongoing clinical trial of FMT to treat MASLD or MASH (ClinicalTrials.gov, as of April 2025). In turn, therapeutic approaches that use specific microorganisms and/or their characterized products have become mainstream in microbial intervention research. Given the potential risk of transmitting uncharacterized or even pathogenic bacteria, it might be safer and better to replace FMT with the administration of probiotics or other known beneficial microorganisms.

Probiotics and engineered bacteria. MASLD can be treated by *Lactobacillus* and *Bifidobacterium*, the two most commonly used probiotics, whilst synergistic administration of multiple probiotics can achieve stronger and more positive effects than a single probiotic strain¹⁸. Two independent RCTs reported that hepatic lipid peroxidation and injury in adult and paediatric patients with MASLD is reduced by a 4-month intervention of a probiotic cocktail, which includes eight lactic acid-producing probiotic strains (three *Bifidobacterium*, four *Lactobacillus*, and *Streptococcus thermophilus*)^{190,191}. Another clinical trial enrolling 39 patients with biopsy-proven MASLD demonstrated that a 1-year intervention of the same probiotic cocktail significantly improved liver histology and reduced serum alanine aminotransferase (ALT) level¹⁹². By contrast, an RCT in 2021 enrolling 35 patients with MASLD reported that 10-week supplementation of the probiotic cocktail fails to relieve liver inflammation and injury, which might be explained by the short treatment duration¹⁹³. Other probiotic formulations,

Table 2 | Clinical trials of microbiome-targeting therapies for MASLD

Trial population	Intervention	Measurement	Primary goals	Outcomes	Limitations	Refs.
FMT						
21 patients with MASLD (15 received allogenic FMT, 6 received autologous FMT)	FMT (3 g of stool) by endoscopy to the distal duodenum	Liver PDFF	Restore gut microbiome	Improved small intestinal permeability but no effect on liver steatosis after 6 months	Small sample size, interobserver variability for MASH diagnosis	185
75 patients with MASLD (47 received FMT, 28 received probiotics)	FMT by colonoscopy	FibroTouch	Improve gut microbial dysbiosis	Reduced fat accumulation in patients with non-obesity MASLD but not in obesity MASLD after 4 weeks	Non-FMT group was treated with probiotics, which is not an appropriate control; long-term efficacy was not evaluated	184
21 patients with MASLD (10 received allogenic FMT from vegan donors, 11 received autologous FMT)	FMT by gastroduodenoscopy	Liver biopsy	Restore gut microbiome	Improved liver histology and downregulated genes related to liver inflammation, lipid metabolism and DNA methylation	Small sample size; baseline ALT and AST levels are different between autologous and allogenic FMT groups	186,187
Probiotics						
22 patients with MASLD (all treated with a probiotic cocktail)	4-month probiotics ^a	Serum liver injury markers	Restore gut microbiome, inhibit inflammation	Improved plasma levels of MDA and 4-HNE	Small sample size, lack of liver histology	190
44 paediatric patients with MASLD (22 probiotics, 22 placebo)	4-month probiotics ^a	Ultrasonography	Restore gut microbiome, inhibit inflammation	Improved MASLD	Small sample size; long-term efficacy was not evaluated	191
39 patients with MASLD (19 probiotics, 20 placebo)	12-month probiotics ^a	Liver biopsy and serum ALT level	Restore gut microbiome, inhibit inflammation	Improved liver histology, serum ALT and cytokines	Small sample size; majority of patients had mild liver histology at baseline	192
35 patients with MASLD (19 probiotics, 16 placebo)	10-week probiotics ^a	Serum ALT and AST levels	Restore gut microbiome, inhibit inflammation	No effect on liver injury biomarkers	Small sample size, short treatment duration	193
20 patients with MASH (10 probiotics, 10 placebo)	6-month Lепicol probiotic formula (<i>Lactobacillus</i> species <i>L. plantarum</i> , <i>L. delbrueckii</i> , <i>L. rhamnosus</i> , <i>L. acidophilus</i> and <i>Bifidobacterium bifidum</i>)	MRS, liver biochemistry and metabolic profile	Restore gut microbiome, inhibit inflammation	Reduced liver fat content and injury	Small sample size; results cannot be directly extrapolated to other probiotic formula	194
28 patients with MASLD (14 probiotics, 14 placebo)	3-month <i>Lactobacillus bulgaricus</i> and <i>Streptococcus thermophilus</i>	Serum liver injury markers	Restore gut microbiome, inhibit inflammation	Reduced serum ALT, AST and γGT levels	Small sample size, lack of liver histology	195
96 patients with MASLD (48 with <i>Clostridium butyricum</i> and rosuvastatin, 48 with rosuvastatin)	6-month <i>C. butyricum</i> with or without rosuvastatin	Ultrasound and blood lipid and liver function	Restore gut microbiome, inhibit inflammation and lipid metabolism	Improved liver function	Control group was treated with rosuvastatin, which may alter the gut microbiome	197
Prebiotics/postbiotics						
200 patients with MASLD	4-month resistant starch type 2	MRS	Alter gut microbiome composition and functionality	Reduced liver fat content and enzymes	Histological changes were not evaluated; potential confounding effect of diet was not controlled	204
32 patients with MASLD	6-month high dietary fibres	Serum liver injury markers	Improve intestinal barrier function	Reduced serum ALT, AST and γGT levels	Small sample size, lack of liver histology	206

Table 2 (continued) | Clinical trials of microbiome-targeting therapies for MASLD

Trial population	Intervention	Measurement	Primary goals	Outcomes	Limitations	Refs.
Prebiotics/postbiotics (continued)						
104 patients with MASLD (55 synbiotic agent, 49 placebo)	12-month synbiotic agent (fructo-oligosaccharides and <i>Bifidobacterium animalis</i>)	MRS	Restore gut microbiome	No effect on liver fat content and fibrosis markers	Synbiotic only included a single strain of probiotic	207
32 patients with obesity or overweight (12 pasteurized <i>Akkermansia muciniphila</i> , 9 living <i>A. muciniphila</i> , 11 placebo)	3-month live or pasteurized <i>A. muciniphila</i>	Serum liver injury markers	Improve insulin resistance, reduce body weight, fat mass and hip circumference	Reduced liver injury and inflammation	Small sample size; physical activity level and calorie intake were not considered	210
Antimicrobials						
50 patients with MASH (25 rifaximin, 25 placebo)	6-month rifaximin	Serum liver injury markers	Eradicate a broad spectrum of gut bacteria	Reduced serum ALT, AST and γ GT levels	Small sample size; second liver biopsy 6 months after the treatment was not performed	213
Microbial-derived metabolites						
185 patients with MASH	18-month UDCA	Liver histology	Inhibit oxidation, inflammation and fibrosis	Improved overall liver histology	Lack of wash-out	215
126 patients with MASH	12-month UDCA	Serum ALT level	Inhibit oxidation, inflammation and fibrosis	Reduced serum ALT level	Lack of liver histology	243

4-HNE, 4-hydroxynonenal; γ GT, γ -glutamyl transferase; ALT, alanine aminotransferase; AST, aspartate aminotransferase; FMT, faecal microbiota transplantation; MASH, metabolic dysfunction-associated steatohepatitis; MASLD, metabolic dysfunction-associated steatotic liver disease; MDA, malondialdehyde; MRS, magnetic resonance spectrometry; PDFF, magnetic resonance imaging-derived proton density fat fraction; UDCA, ursodeoxycholic acid. ^aIncludes eight lactic acid-producing probiotic strains (three *Bifidobacterium*, four *Lactobacillus*, and *Streptococcus thermophilus*).

such as the commercially available cocktail Lepicol (which consists of four *Lactobacillus* and one *Bifidobacterium*), also showed positive results by reducing liver fat content and injury in a proof-of-concept RCT involving 20 patients with MASH¹⁹⁴. Three-month intervention of *Lactobacillus bulgaricus* and *S. thermophilus* in 28 patients with MASLD significantly improves liver ALT level¹⁹⁵. Two meta-analyses, enrolling 28 and 9 clinical trials, respectively, have confirmed the efficacy of probiotics to treat MASLD, showing improved liver function and reduced blood glucose levels¹⁸. Notably, most of the aforementioned studies were conducted with a relatively small sample size, raising questions about their robustness and replicability. Moreover, some murine studies reported that carcinogen-induced and diet-induced MASLD-related HCC could be suppressed by probiotics administration (for example, *B. pseudolongum* and *L. acidophilus*)^{114,115}, yet the potential of probiotics to treat liver malignancy in patients remains uncertain.

Owing to the rapid advances in microbial profiling technology, beneficial bacteria that commensally live in the human gut microbiome can be identified. After thoroughly understanding their functional role and mechanism of action, these newly identified bacteria can also be utilized as probiotics, often referred to as ‘next-generation probiotics’. Unlike conventional probiotics, these new strains offer benefits, such as greater disease-targeting precision, improved safety profile and easier colonization, as they are natural inhabitants of the human gut. Of these next-generation probiotics, *A. muciniphila* seems to be more related to obesity and MASLD as a clinical trial reported that the faecal abundance of *A. muciniphila* is associated with better clinical outcomes after calorie restriction in adults with overweight or obesity¹⁹⁶. A 6-month supplementation of *Clostridium butyricum* significantly reduced blood lipid levels and alleviated liver fibrosis and injury in a study enrolling

98 patients with MASLD¹⁹⁷. *P. distasonis*, when provided with dietary inulin as a substrate, offers protection against MASH development in mice by producing the anti-steatotic metabolite pentadecanoic acid⁷⁴. Although bacterial probiotics have been the primary focus of research, several preclinical studies also highlight the potential of yeast probiotics to treat MASH, particularly *Saccharomyces boulardii*. *S. boulardii* supplementation leads to reduced liver steatosis, injury, inflammation and fibrosis in transgenic obese (*db/db*) mice¹⁹⁸ and mice with diet-induced MASH¹⁹⁹. Nonetheless, most clinical studies of *S. boulardii* focused on inflammatory bowel disease²⁰⁰, thereby leaving a gap for future studies to evaluate the efficacy of fungal probiotics in the treatment of liver diseases.

Engineered bacteria represent another frontier in microbiome-targeting therapies²⁰¹. They were designed to produce beneficial metabolites or metabolize toxins into non-toxin products. For example, genetically modified IL-22-producing *Lactobacillus reuteri* has shown potential to suppress steatohepatitis in mice²⁰². However, as a newly emerging field, the use of engineered bacterial strains as therapeutics requires careful consideration and extensive research.

Prebiotics and postbiotics. Prebiotics are non-digestible food substances that nourish gut commensal probiotics to aid their growth and the production of beneficial compounds. Preclinical studies have revealed that supplementation of prebiotics such as starch, inulin and fructo-oligosaccharides could alleviate MASH in mice^{74,203}. An RCT involving 200 patients with MASLD reported that 4-month supplementation of resistant starch from a prebiotic high-amylose maize induces a significant reduction of liver fat content and serum lipid level concomitant with improved liver enzymes and insulin sensitivity^{204,205}. Another

RCT enrolling 32 patients with MASLD showed that a high intake of dietary fibres for 6 months reduces liver enzymes and improves liver steatosis, suggesting that increased dietary fibre intake should be recommended for disease management in patients with MASLD²⁰⁶. By contrast, a double-blind phase II trial involving 104 patients with MASLD found that 1-year administration of a synbiotic agent (combining prebiotic fructo-oligosaccharides and probiotic *Bifidobacterium animalis* subspecies lactis BB-12) alters the gut microbiome but fails to reduce liver fat or improve liver fibrosis²⁰⁷.

Postbiotics are defined as the preparation of inactive microorganisms or their components that confer health benefits to the host²⁰⁸. A preclinical study suggested that pasteurized *A. muciniphila* provides even stronger beneficial effects than its live form in mice and humans with obesity by interacting with TLR2 using its outer membrane protein Amuc_1100 (ref. 209). A subsequent RCT in insulin-resistant patients with obesity involving 3-month supplementation of *A. muciniphila* confirmed the safety of both live and pasteurized *A. muciniphila* in humans²¹⁰. The pasteurized *A. muciniphila* significantly improves insulin sensitivity, increases weight loss, alleviates endotoxaemia, and suppresses liver injury and inflammation in patients compared with the placebo group.

Antimicrobials. Treatment with a combination of antibiotics has demonstrated an ability to blunt MASLD progression in preclinical studies without compromising intestinal barrier integrity²¹¹. Rifaximin, a broad-spectrum antibiotic, suppresses MASLD in mice by altering the gut microbiome to lower ileal DCA levels²¹². In a RCT enrolling 50 patients with biopsy-proven MASH, a 6-month treatment course with rifaximin results in decreased levels of ALT, AST, γ -glutamyl transferase and pro-inflammatory cytokines²¹³. Furthermore, a preclinical study illustrated that antifungal treatment with amphotericin B could dampen the development of diet-induced MASH in gnotobiotic mice transplanted with stools from patients with MASH⁴⁵.

Microbial-derived metabolites. In certain situations, especially for immunocompromised individuals, the use of microorganism-derived products might be safer and more beneficial than live bacteria. For example, UDCA is a secondary bile acid that suppresses MASLD in rats by activating AMPK to inhibit apoptosis and improve autophagy²¹⁴. Numerous RCTs have verified the efficacy of UDCA to treat MASLD²¹⁵. For example, two meta-analyses revealed that UDCA effectively reduces the levels of the liver damage markers ALT and AST in patients with MASLD^{216,217}. Another bile acid, HDCA, alleviates MASLD in multiple mouse models by inhibiting intestinal FXR and promoting the growth of probiotic *P. distasonis* in the gut¹⁰³. SCFAs are another class of metabolites that suppress MASLD by reducing cholesterol and inducing autophagy and epigenetic modulation²¹⁸. For example, butyrate has been shown to alter microbiome composition, improve intestinal barrier integrity and reduce serum endotoxins, thereby ameliorating MASLD progression in mice²¹⁹. Other SCFAs, including acetate and valerate, could suppress MASLD-related HCC development by downregulating the JAK–STAT3 and Rho–GTPase pathways, respectively^{114,115}. Notably, despite these promising preclinical findings, clinical research on bile acids and SCFAs as therapeutics for MASLD remains limited, primarily due to challenges in controlling treatment dosage and duration²¹⁸. Thus, further clinical research is needed to overcome current limitations and establish effective and optimized treatment protocols.

Bacteriophages. Bacteriophages (also known as phages) compose 90% of the human gut virome and can precisely edit microbiome composition

by killing and eliminating the target bacteria. Preclinical studies have shown that phages specifically targeting ethanol-producing *K. pneumoniae* suppress MASLD in mice²²⁰. A clinical trial involving 18 healthy volunteers demonstrated that a combination of five phages targeting *K. pneumoniae* is safe and viable for human use²²¹. However, using phages to treat MASLD is still at a preliminary stage, and several challenges have hindered clinical application. For example, phages are highly specific and can only lyse limited bacterial strains²²². Pharmacokinetics and pharmacodynamics of phages in the human body also require careful consideration. More importantly, bacteria can develop different resistance mechanisms to avoid phage invasion, such as restriction-modification, CRISPR–Cas adaptive immunity, surface receptor modification and using receptors as phage decoys²²³, hence greatly reducing the efficacy of phage therapy. To manage phage resistance, one strategy is to combine multiple phages in cocktails to broaden the target range of bacterial hosts²²⁴. Another approach is to use antibiotics synergistically with phages to prevent the emergence of phage-resistant bacteria^{225,226}. Notably, although phage cocktails and antibiotics significantly reduce the frequency of phage-resistant mutations in preclinical studies, neither of these strategies has shown robust therapeutic improvement in humans²²⁷.

Current challenges and future directions

Basic research

Data reproducibility has been a major issue in microbiome research as illustrated by the inconsistent microbial alterations during MASLD progression (Table 1). The differences in cohort design and various environmental factors including age, sex and geographic region have greatly contributed to data inconsistency across studies. The disparity in sample processing (for example, DNA extraction protocols, DNA amplification by PCR prior to library preparation and host DNA contamination), sequencing platform and depth (for example, second-generation or third-generation sequencers), and analysis pipelines can further aggregate such variations^{228,229}. To this end, large-scale multi-population studies with validation cohorts are suggested to ensure the findings can be replicated. Meta-analysis, which combines results from comparable studies after adjusting for clinical variables and other confounding factors, yields improved statistical power and enables the reduction of cohort-specific bias²³⁰. Thus, it is also a robust approach for identifying key MASLD-associated microorganisms that are consistently altered in multiple metagenomic studies.

While gut microbial dysbiosis in MASLD and related conditions has been extensively profiled (albeit only for bacteria – alterations in gut viruses and fungi remain hugely unclear), the underlying mechanisms are yet to be addressed. Microbiome studies in intestinal diseases, such as inflammatory bowel disease and colorectal cancer, have conducted in-depth mechanistic investigations, which not only validate *in silico* findings from sequencing but also elucidate the mechanistic insights of the key microorganisms. By comparison, how enriched (for example, *Clostridium* and *Streptococcus*) or depleted (for example, *Eubacterium* and *Faecalibacterium*) bacteria respectively promote or inhibit MASLD progression as well as their influence on host immunity, genetics and epigenetics, remain unknown. Meanwhile, multi-omics analysis is fundamental for investigating host–microorganism interactions through coupling microbial profiling with other omics techniques, including metabolomics, transcriptomics, epigenomics and proteomics. In addition, tracking the translocation of gut microorganisms and their derivatives and metabolites in the gut–liver axis has been challenging. Fortunately, novel technologies, such as organ-on-chip models, might

provide valuable insights into these underlying mechanisms²³¹. Furthermore, given the importance of diet in MASLD, tracking gut microorganisms that are responsible for metabolizing specific dietary components is also important and can be achieved by isotope probing or labelling⁷⁴.

Translational research

Increased understanding of the gut microbiome has facilitated clinical application of gut microorganisms and/or metabolites as biomarkers²³². However, although these microbial biomarkers represent an exciting development in noninvasive diagnosis for MASLD and related diseases, it is pivotal to validate their performance and scoring system in independent cohorts across diverse geographical, dietary, ethnic and environmental backgrounds. Comparative studies are also necessary to assess the performance of these microbiome-based biomarkers against existing clinical and imaging-based methods. An important area for future research is the development of noninvasive tests to assess the risk of clinical outcomes, including progression to cirrhosis, hepatic decompensation and mortality. Moreover, longitudinal studies are indispensable to assess the reliability of microbial biomarkers. In the era of precision medicine, microbiome-based biomarkers have the potential to more accurately identify individuals at increased risk of liver diseases, enabling more targeted and effective interventions.

The gut microbiome offers new opportunities for the development of microbiome-targeting approaches (for example, probiotics and prebiotics/postbiotics) for MASLD prevention and treatment. Nonetheless, while these pioneering studies have shown promising potential, there are many unsolved issues prior to their clinical application, including intervention duration, optimal dosage and efficacy. Current evidence has highlighted the great diversity in therapeutic outcomes of microbiome-targeting approaches among individuals. To this end, it is worth investigating the synergistic effects of incorporating different approaches, for example, combining multiple next-generation probiotics together or with other traditional probiotics. Moreover, as it is unlikely that all patients with MASLD have the same alterations in their gut microbiome, personalized treatment approaches that address specific microbial strains might enhance the efficacy of medical interventions for MASLD and related conditions. On the other hand, increasing studies have indicated that gut transit time (referring to the time it takes foods to travel through the human digestive tract) is the main determinant that explains the inter-individual variations in microbiome diversity, composition and metabolism^{233–235}. For example, a long gut transit time is associated with a shift of microbiome-mediated metabolism from saccharolysis towards proteolysis, resulting in SCFA reduction^{236,237}. Hence, to reduce inter-individual variation and improve efficacy, gut transit time should be considered as a confounding factor in microbial intervention studies²³⁶. In addition, another uncharted area of MASLD-related HCC research is the use of the microbiome as an adjuvant for enhancing the efficacy of systemic therapy. To date, accumulated evidence has shown that targeting the gut microbiome by FMT or probiotics could markedly improve the efficacy of chemotherapy or immunotherapy against different cancers^{238–240} albeit not MASLD-related HCC. As patients with MASLD-related HCC often show poor response to immune-checkpoint blockade¹⁷², identifying probiotics or engineered bacteria that boost immunotherapy efficacy might provide useful adjuvants to enhance treatment outcomes.

Conclusions

Extensive research has revealed the role of gut microbial dysbiosis in the progression from MASLD, to MASH, to MASLD-related fibrosis and

finally to HCC. This not only involves bacteria and their metabolites but also fungi and viruses. This information offers the potential for noninvasive diagnostic biomarkers to identify individuals who are pre-disposed to increased risk of disease progression. However, it is crucial to conduct longitudinal studies, across diverse ethnic backgrounds, using multiple independent cohorts to confirm the reliability of these microbial biomarkers. Gaining a comprehensive understanding of the gut microbiome in MASLD and MASLD-related HCC creates new opportunities for the development of innovative drugs that specifically target gut microorganisms. As it is unlikely that all patients with MASLD have the same dysbiosis in their gut microbiome, personalized treatment approaches that address specific microbial strains would enhance the efficacy of medical interventions for MASLD and related diseases.

Published online: 7 July 2025

References

1. Tripathi, A. et al. The gut–liver axis and the intersection with the microbiome. *Nat. Rev. Gastroenterol. Hepatol.* **15**, 397–411 (2018).
2. Albillos, A., de Gottardi, A. & Rescigno, M. The gut–liver axis in liver disease: pathophysiological basis for therapy. *J. Hepatol.* **72**, 558–577 (2020).
3. Hsu, C. L. & Schnabl, B. The gut–liver axis and gut microbiota in health and liver disease. *Nat. Rev. Microbiol.* **21**, 719–733 (2023).
4. Powell, E. E., Wong, V. W. & Rinella, M. Non-alcoholic fatty liver disease. *Lancet* **397**, 2212–2224 (2021).
5. Singh, S. et al. Fibrosis progression in nonalcoholic fatty liver vs nonalcoholic steatohepatitis: a systematic review and meta-analysis of paired-biopsy studies. *Clin. Gastroenterol. Hepatol.* **13**, 643–654.e9 (2015).
6. Chalasani, N. et al. The diagnosis and management of nonalcoholic fatty liver disease: practice guidance from the American Association for the Study of Liver Diseases. *Hepatology* **67**, 328–357 (2018).
7. Raman, M. et al. Fecal microbiome and volatile organic compound metabolome in obese humans with nonalcoholic fatty liver disease. *Clin. Gastroenterol. Hepatol.* **11**, 868–875.e3 (2013).
8. Wong, V. W. et al. Molecular characterization of the fecal microbiota in patients with nonalcoholic steatohepatitis — a longitudinal study. *PLoS ONE* **8**, e62885 (2013).
9. Shen, F. et al. Gut microbiota dysbiosis in patients with non-alcoholic fatty liver disease. *Hepatobiliary Pancreat. Dis. Int.* **16**, 375–381 (2017).
10. Zhu, L. et al. Characterization of gut microbiomes in nonalcoholic steatohepatitis (NASH) patients: a connection between endogenous alcohol and NASH. *Hepatology* **57**, 601–609 (2013).
11. Aron-Wisniewsky, J. et al. Gut microbiota and human NAFLD: disentangling microbial signatures from metabolic disorders. *Nat. Rev. Gastroenterol. Hepatol.* **17**, 279–297 (2020).
12. Lee, G. et al. Distinct signatures of gut microbiome and metabolites associated with significant fibrosis in non-obese NAFLD. *Nat. Commun.* **11**, 4982 (2020).
13. Loomba, R. et al. Gut microbiome-based metagenomic signature for non-invasive detection of advanced fibrosis in human nonalcoholic fatty liver disease. *Cell Metab.* **25**, 1054–1062.e5 (2017).
14. Hoyle, L. et al. Molecular phenomics and metagenomics of hepatic steatosis in non-diabetic obese women. *Nat. Med.* **24**, 1070–1080 (2018).
15. Kwan, S. Y. et al. Gut microbiome features associated with liver fibrosis in Hispanics, a population at high risk for fatty liver disease. *Hepatology* **75**, 955–967 (2022).
16. Carr, R. M. et al. An integrated analysis of fecal microbiome and metabolomic features distinguish non-cirrhotic NASH from healthy control populations. *Hepatology* **78**, 1843–1857 (2023).
17. Schwimmer, J. B. et al. Microbiome signatures associated with steatohepatitis and moderate to severe fibrosis in children with nonalcoholic fatty liver disease. *Gastroenterology* **157**, 1109–1122 (2019).
18. Yang, R. et al. Effects of probiotics on nonalcoholic fatty liver disease: a systematic review and meta-analysis. *Expert Rev. Gastroenterol. Hepatol.* **15**, 1401–1409 (2021).
19. Zeng, X. et al. Pien Tze Huang protects against non-alcoholic steatohepatitis by modulating the gut microbiota and metabolites in mice. *Engineering* **35**, 257–269 (2024).
20. Kajikawa, A. et al. Characterization of flagellins isolated from a highly motile strain of *Lactobacillus agilis*. *BMC Microbiol.* **16**, 49 (2016).
21. Kim, D. & Kim, W. R. Nonobese fatty liver disease. *Clin. Gastroenterol. Hepatol.* **15**, 474–485 (2017).
22. Leung, J. C. et al. Histological severity and clinical outcomes of nonalcoholic fatty liver disease in nonobese patients. *Hepatology* **65**, 54–64 (2017).
23. Duarte, S. M. B. et al. Gut microbiome composition in lean patients with NASH is associated with liver damage independent of caloric intake: a prospective pilot study. *Nutr. Metab. Cardiovasc. Dis.* **28**, 369–384 (2018).
24. Iwaki, M. et al. Gut microbiota composition associated with hepatic fibrosis in non-obese patients with non-alcoholic fatty liver disease. *J. Gastroenterol. Hepatol.* **36**, 2275–2284 (2021).

25. Kuchay, M. S., Martinez-Montoro, J. I., Choudhary, N. S., Fernandez-Garcia, J. C. & Ramos-Molina, B. Non-alcoholic fatty liver disease in lean and non-obese individuals: current and future challenges. *Biomedicines* **9**, 1346 (2021).
26. Ha, Y. et al. Intimate association of visceral obesity with non-alcoholic fatty liver disease in healthy Asians: a case-control study. *J. Gastroenterol. Hepatol.* **30**, 1666–1672 (2015).
27. Feng, R. N. et al. Lean non-alcoholic fatty liver disease increases risk for metabolic disorders in a normal weight Chinese population. *World J. Gastroenterol.* **20**, 17932–17940 (2014).
28. Qin, N. et al. Alterations of the human gut microbiome in liver cirrhosis. *Nature* **513**, 59–64 (2014).
29. Ponzi, F. R. et al. Hepatocellular carcinoma is associated with gut microbiota profile and inflammation in nonalcoholic fatty liver disease. *Hepatology* **69**, 107–120 (2019).
30. Caussy, C. et al. A gut microbiome signature for cirrhosis due to nonalcoholic fatty liver disease. *Nat. Commun.* **10**, 1406 (2019).
31. Oh, T. G. et al. A universal gut-microbiome-derived signature predicts cirrhosis. *Cell Metab.* **32**, 878–888.e6 (2020).
32. Behary, J. et al. Gut microbiota impact on the peripheral immune response in non-alcoholic fatty liver disease related hepatocellular carcinoma. *Nat. Commun.* **12**, 187 (2021).
33. Quevrain, E. et al. Identification of an anti-inflammatory protein from *Faecalibacterium prausnitzii*, a commensal bacterium deficient in Crohn's disease. *Gut* **65**, 415–425 (2016).
34. Lu, H. et al. Butyrate-producing *Eubacterium rectale* suppresses lymphomagenesis by alleviating the TNF-induced TLR4/MyD88/NF- κ B axis. *Cell Host Microbe* **30**, 1139–1150.e7 (2022).
35. Atarashi, K. et al. Treg induction by a rationally selected mixture of *Clostridia* strains from the human microbiota. *Nature* **500**, 232–236 (2013).
36. Narushima, S. et al. Characterization of the 17 strains of regulatory T cell-inducing human-derived *Clostridia*. *Gut Microbes* **5**, 333–339 (2014).
37. Scheiman, J. et al. Meta-omics analysis of elite athletes identifies a performance-enhancing microbe that functions via lactate metabolism. *Nat. Med.* **25**, 1104–1109 (2019).
38. Leung, H. et al. Risk assessment with gut microbiome and metabolite markers in NAFLD development. *Sci. Transl. Med.* **14**, eabk0855 (2022).
39. Rothschild, D. et al. Environment dominates over host genetics in shaping human gut microbiota. *Nature* **555**, 210–215 (2018).
40. Yatsunenko, T. et al. Human gut microbiome viewed across age and geography. *Nature* **468**, 222–227 (2012).
41. Qin, J. et al. A human gut microbial gene catalogue established by metagenomic sequencing. *Nature* **464**, 59–65 (2010).
42. Arumugam, M. et al. Enterotypes of the human gut microbiome. *Nature* **473**, 174–180 (2011).
43. Underhill, D. M. & Iliev, I. D. The mycobiota: interactions between commensal fungi and the host immune system. *Nat. Rev. Immunol.* **14**, 405–416 (2014).
44. You, N. et al. Fecal fungi dysbiosis in nonalcoholic fatty liver disease. *Obesity* **29**, 350–358 (2021).
45. Demir, M. et al. The fecal mycobiome in non-alcoholic fatty liver disease. *J. Hepatol.* **76**, 788–799 (2022).
46. Niu, C. et al. Mapping the human oral and gut fungal microbiota in patients with metabolic dysfunction-associated fatty liver disease. *Front. Cell Infect. Microbiol.* **13**, 1157368 (2023).
47. Viebahn, G. et al. Fungal signature differentiates alcohol-associated liver disease from nonalcoholic fatty liver disease. *Gut Microbes* **16**, 2307586 (2024).
48. Pak, J., Tucci, V. T., Vincent, A. L., Sandin, R. L. & Greene, J. N. Mucormycosis in immunochallenged patients. *J. Emerg. Trauma Shock* **1**, 106–113 (2008).
49. Lopes, J. P. & Lionakis, M. S. Pathogenesis and virulence of *Candida albicans*. *Virulence* **13**, 89–121 (2022).
50. Guerin, E. et al. Biology and taxonomy of crass-like bacteriophages, the most abundant virus in the human gut. *Cell Host Microbe* **24**, 653–664.e6 (2018).
51. Lang, S. et al. Intestinal virome signature associated with severity of nonalcoholic fatty liver disease. *Gastroenterology* **159**, 1839–1852 (2020).
52. Strathdee, S. A., Hatfull, G. F., Mutalik, V. K. & Schooley, R. T. Phage therapy: from biological mechanisms to future directions. *Cell* **186**, 17–31 (2023).
53. Manrique, P. et al. Healthy human gut phageome. *Proc. Natl Acad. Sci. USA* **113**, 10400–10405 (2016).
54. Shkoporov, A. N. et al. The human gut virome is highly diverse, stable, and individual specific. *Cell Host Microbe* **26**, 527–541.e5 (2019).
55. Shkoporov, A. N. & Hill, C. Bacteriophages of the human gut: the “Known Unknown” of the microbiome. *Cell Host Microbe* **25**, 195–209 (2019).
56. Aggarwala, V., Liang, G. & Bushman, F. D. Viral communities of the human gut: metagenomic analysis of composition and dynamics. *Mob. DNA* **8**, 12 (2017).
57. Leinwand, J. C. et al. Intrahepatic microbes govern liver immunity by programming NKT cells. *J. Clin. Invest.* **132**, e151725 (2022).
58. Sookoian, S. et al. Intrahepatic bacterial metatranscriptomic signature in non-alcoholic fatty liver disease. *Gut* **69**, 1483–1491 (2020).
59. Liu, B. et al. Hepatic stellate cell activation and senescence induced by intrahepatic microbiota disturbances drive progression of liver cirrhosis toward hepatocellular carcinoma. *J. Immunother. Cancer* **10**, e003069 (2022).
60. Sun, L. et al. Intratumoral microbiome can predict the prognosis of hepatocellular carcinoma after surgery. *Clin. Transl. Med.* **13**, e1331 (2023).
61. Wang, X. et al. Gut-liver translocation of pathogen *Klebsiella pneumoniae* promotes hepatocellular carcinoma in mice. *Nat. Microbiol.* **10**, 169–184 (2025).
62. Huang, J. H. et al. The intratumoral bacterial metatranscriptomic signature of hepatocellular carcinoma. *Microbiol. Spectr.* **10**, e0098322 (2022).
63. He, Y., Zhang, Q., Yu, X., Zhang, S. & Guo, W. Overview of microbial profiles in human hepatocellular carcinoma and adjacent nontumor tissues. *J. Transl. Med.* **21**, 68 (2023).
64. Li, S. et al. Intratumoral microbial heterogeneity affected tumor immune microenvironment and determined clinical outcome of HBV-related HCC. *Hepatology* **78**, 1079–1091 (2023).
65. Effenberger, M. et al. A gut bacterial signature in blood and liver tissue characterizes cirrhosis and hepatocellular carcinoma. *Hepatology. Commun.* **7**, e00182 (2023).
66. Guo, W. et al. Diet and risk of non-alcoholic fatty liver disease, cirrhosis, and liver cancer: a large prospective cohort study in UK Biobank. *Nutrients* **14**, 5335 (2022).
67. Petermann-Rocha, F. et al. Associations between an inflammatory diet index and severe non-alcoholic fatty liver disease: a prospective study of 171,544 UK Biobank participants. *BMC Med.* **21**, 123 (2023).
68. Loomba, R., Friedman, S. L. & Shulman, G. I. Mechanisms and disease consequences of nonalcoholic fatty liver disease. *Cell* **184**, 2537–2564 (2021).
69. European Association for the Study of the Liver; European Association for the Study of Diabetes; European Association for the Study of Obesity. EASL-EASD-EASO clinical practice guidelines on the management of metabolic dysfunction-associated steatotic liver disease (MASLD). *J. Hepatol.* **81**, 492–542 (2024).
70. George, E. S. et al. The association between diet and hepatocellular carcinoma: a systematic review. *Nutrients* **13**, 172 (2021).
71. Yaskolka Meir, A. et al. Effect of green-Mediterranean diet on intrahepatic fat: the DIRECT PLUS randomised controlled trial. *Gut* **70**, 2085–2095 (2021).
72. Vilar-Gomez, E. et al. High-quality diet, physical activity, and college education are associated with low risk of NAFLD among the US population. *Hepatology* **75**, 1491–1506 (2022).
73. Zhang, X. et al. Dietary cholesterol drives fatty liver-associated liver cancer by modulating gut microbiota and metabolites. *Gut* **70**, 761–774 (2021).
74. Wei, W. et al. *Parabacteroides distasonis* uses dietary inulin to suppress NASH via its metabolite pentadecanoic acid. *Nat. Microbiol.* **8**, 1534–1548 (2023).
75. Lange, N. F., Radu, P. & Dufour, J. F. Prevention of NAFLD-associated HCC: role of lifestyle and chemoprevention. *J. Hepatol.* **75**, 1217–1227 (2021).
76. Chen, B. et al. Gut bacteria alleviate smoking-related NASH by degrading gut nicotine. *Nature* **610**, 562–568 (2022).
77. Zhang, X., Chang, K. M., Yu, J. & Loomba, R. Unraveling mechanisms of genetic risks in metabolic dysfunction-associated steatotic liver diseases: a pathway to precision medicine. *Annu. Rev. Pathol.* **20**, 375–403 (2025).
78. Ha, S., Wong, V. W., Zhang, X. & Yu, J. Interplay between gut microbiome, host genetic and epigenetic modifications in MASLD and MASLD-related hepatocellular carcinoma. *Gut* **74**, 141–152 (2025).
79. Zhang, X. et al. Intestinal TM6SF2 protects against metabolic dysfunction-associated steatohepatitis through the gut-liver axis. *Nat. Metab.* **7**, 102–119 (2025).
80. Harte, A. L. et al. Elevated endotoxin levels in non-alcoholic fatty liver disease. *J. Inflamm.* **7**, 15 (2010).
81. Verdam, F. J., Rensen, S. S., Driessen, A., Greve, J. W. & Buurman, W. A. Novel evidence for chronic exposure to endotoxin in human nonalcoholic steatohepatitis. *J. Clin. Gastroenterol.* **45**, 149–152 (2011).
82. Giorgio, V. et al. Intestinal permeability is increased in children with non-alcoholic fatty liver disease, and correlates with liver disease severity. *Dig. Liver Dis.* **46**, 556–560 (2014).
83. Ruiz, A. G. et al. Lipopolysaccharide-binding protein plasma levels and liver TNF- α gene expression in obese patients: evidence for the potential role of endotoxin in the pathogenesis of non-alcoholic steatohepatitis. *Obes. Surg.* **17**, 1374–1380 (2007).
84. Miele, L. et al. Increased intestinal permeability and tight junction alterations in nonalcoholic fatty liver disease. *Hepatology* **49**, 1877–1887 (2009).
85. Xin, D., Zong-Shun, L., Bang-Mao, W. & Lu, Z. Expression of intestinal tight junction proteins in patients with non-alcoholic fatty liver disease. *Hepatogastroenterology* **61**, 136–140 (2014).
86. Rahman, K. et al. Loss of junctional adhesion molecule promotes severe steatohepatitis in mice on a diet high in saturated fat, fructose, and cholesterol. *Gastroenterology* **151**, 733–746.e12 (2016).
87. Mouries, J. et al. Microbiota-driven gut vascular barrier disruption is a prerequisite for non-alcoholic steatohepatitis development. *J. Hepatol.* **71**, 1216–1228 (2019).
88. Feng, Y., Wang, Y., Wang, P., Huang, Y. & Wang, F. Short-chain fatty acids manifest stimulative and protective effects on intestinal barrier function through the inhibition of NLRP3 inflammasome and autophagy. *Cell Physiol. Biochem.* **49**, 190–205 (2018).
89. Singh, R. et al. Enhancement of the gut barrier integrity by a microbial metabolite through the Nrf2 pathway. *Nat. Commun.* **10**, 89 (2019).
90. Ghosh, S., Whitley, C. S., Haribabu, B. & Jala, V. R. Regulation of intestinal barrier function by microbial metabolites. *Cell Mol. Gastroenterol. Hepatol.* **11**, 1463–1482 (2021).
91. Ravcheev, D. A. & Thiele, I. Systematic genomic analysis reveals the complementary aerobic and anaerobic respiration capacities of the human gut microbiota. *Front. Microbiol.* **5**, 674 (2014).
92. Patel, D. & Witt, S. N. Ethanolamine and phosphatidylethanolamine: partners in health and disease. *Oxid. Med. Cell Longev.* **2017**, 4829180 (2017).
93. Mishra, S. P. et al. A mechanism by which gut microbiota elevates permeability and inflammation in obese/diabetic mice and human gut. *Gut* **72**, 1848–1865 (2023).
94. Jiao, N. et al. Suppressed hepatic bile acid signalling despite elevated production of primary and secondary bile acids in NAFLD. *Gut* **67**, 1881–1891 (2018).
95. Smirnova, E. et al. Metabolic reprogramming of the intestinal microbiome with functional bile acid changes underlie the development of NAFLD. *Hepatology* **76**, 1811–1824 (2022).

96. Yoshimoto, S. et al. Obesity-induced gut microbial metabolite promotes liver cancer through senescence secretome. *Nature* **499**, 97–101 (2013).
97. Xie, G. et al. Dysregulated hepatic bile acids collaboratively promote liver carcinogenesis. *Int. J. Cancer* **139**, 1764–1775 (2016).
98. Jia, W., Xie, G. & Jia, W. Bile acid-microbiota crosstalk in gastrointestinal inflammation and carcinogenesis. *Nat. Rev. Gastroenterol. Hepatol.* **15**, 111–128 (2018).
99. Loo, T. M. et al. Gut microbiota promotes obesity-associated liver cancer through PGE₂-mediated suppression of antitumor immunity. *Cancer Discov.* **7**, 522–538 (2017).
100. Liu, L. et al. Deoxycholic acid disrupts the intestinal mucosal barrier and promotes intestinal tumorigenesis. *Food Funct.* **9**, 5588–5597 (2018).
101. Zhou, H. et al. Bile acid toxicity in Paneth cells contributes to gut dysbiosis induced by high-fat feeding. *JCI Insight* **5**, e138881 (2020).
102. Ma, C. et al. Gut microbiome-mediated bile acid metabolism regulates liver cancer via NKT cells. *Science* **360**, eaan5931 (2018).
103. Kuang, J. et al. Hyodeoxycholic acid alleviates non-alcoholic fatty liver disease through modulating the gut-liver axis. *Cell Metab.* **35**, 1752–1766.e8 (2023).
104. Nie, Q. et al. Gut symbionts alleviate MASH through a secondary bile acid biosynthetic pathway. *Cell* **187**, 2717–2734.e33 (2024).
105. Canfora, E. E., Meex, R. C. R., Venema, K. & Blaak, E. E. Gut microbial metabolites in obesity, NAFLD and T2DM. *Nat. Rev. Endocrinol.* **15**, 261–273 (2019).
106. Chambers, E. S. et al. Effects of targeted delivery of propionate to the human colon on appetite regulation, body weight maintenance and adiposity in overweight adults. *Gut* **64**, 1744–1754 (2015).
107. Aoki, R. et al. Commensal microbe-derived acetate suppresses NAFLD/NASH development via hepatic FFAR2 signalling in mice. *Microbiome* **9**, 188 (2021).
108. Yang, X. et al. Inulin-enriched Megamonas funiformis ameliorates metabolic dysfunction-associated fatty liver disease by producing propionic acid. *npj Biofilms Microbiomes* **9**, 84 (2023).
109. Gao, Z. et al. Butyrate improves insulin sensitivity and increases energy expenditure in mice. *Diabetes* **58**, 1509–1517 (2009).
110. Zhao, Z. H. et al. Sodium butyrate supplementation inhibits hepatic steatosis by stimulating liver kinase B1 and insulin-induced gene. *Cell Mol. Gastroenterol. Hepatol.* **12**, 857–871 (2021).
111. Xia, J. K. et al. Roles and regulation of histone acetylation in hepatocellular carcinoma. *Front. Genet.* **13**, 982222 (2022).
112. Olaniyi, K. S. & Amusa, O. A. Sodium acetate-mediated inhibition of histone deacetylase alleviates hepatic lipid dysregulation and its accompanied injury in streptozotocin-nicotinamide-induced diabetic rats. *Biomed. Pharmacother.* **128**, 110226 (2020).
113. Sarkar, A. et al. Butyrate limits inflammatory macrophage niche in NASH. *Cell Death Dis.* **14**, 332 (2023).
114. Song, Q. et al. *Bifidobacterium pseudolongum*-generated acetate suppresses non-alcoholic fatty liver disease-associated hepatocellular carcinoma. *J. Hepatol.* **79**, 1352–1365 (2023).
115. Lau, H. C. et al. *Lactobacillus acidophilus* suppresses non-alcoholic fatty liver disease-associated hepatocellular carcinoma through producing valeric acid. *EBioMedicine* **100**, 104952 (2024).
116. Kim, M. H., Kang, S. G., Park, J. H., Yanagisawa, M. & Kim, C. H. Short-chain fatty acids activate GPR41 and GPR43 on intestinal epithelial cells to promote inflammatory responses in mice. *Gastroenterology* **145**, 396–406.e10 (2013).
117. Belcheva, A. et al. Gut microbial metabolism drives transformation of MSH2-deficient colon epithelial cells. *Cell* **158**, 288–299 (2014).
118. Singh, V. et al. Microbiota-dependent hepatic lipogenesis mediated by stearoyl CoA desaturase 1 (SCD1) promotes metabolic syndrome in TLR5-deficient mice. *Cell Metab.* **22**, 983–996 (2015).
119. Singh, V. et al. Dysregulated microbial fermentation of soluble fiber induces cholestatic liver cancer. *Cell* **175**, 679–694.e22 (2018).
120. Samuel, B. S. et al. Effects of the gut microbiota on host adiposity are modulated by the short-chain fatty-acid binding G protein-coupled receptor, Gpr41. *Proc. Natl Acad. Sci. USA* **105**, 16767–16772 (2008).
121. Bjursell, M. et al. Improved glucose control and reduced body fat mass in free fatty acid receptor 2-deficient mice fed a high-fat diet. *Am. J. Physiol. Endocrinol. Metab.* **300**, E211–E220 (2011).
122. Chu, H., Duan, Y., Yang, L. & Schnabl, B. Small metabolites, possible big changes: a microbiota-centered view of non-alcoholic fatty liver disease. *Gut* **68**, 359–370 (2019).
123. Rau, M. et al. Fecal SCFAs and SCFA-producing bacteria in gut microbiome of human NAFLD as a putative link to systemic T-cell activation and advanced disease. *United European Gastroenterol. J.* **6**, 1496–1507 (2018).
124. Michail, S. et al. Altered gut microbial energy and metabolism in children with non-alcoholic fatty liver disease. *FEMS Microbiol. Ecol.* **91**, 1–9 (2015).
125. Deng, K. et al. Comparison of fecal and blood metabolome reveals inconsistent associations of the gut microbiota with cardiometabolic diseases. *Nat. Commun.* **14**, 571 (2023).
126. Badawy, A. A. Kynurenine pathway of tryptophan metabolism: regulatory and functional aspects. *Int. J. Tryptophan Res.* **10**, 1178646917691938 (2017).
127. Agus, A., Planchais, J. & Sokol, H. Gut microbiota regulation of tryptophan metabolism in health and disease. *Cell Host Microbe* **23**, 716–724 (2018).
128. Sui, G., Jia, L., Quan, D., Zhao, N. & Yang, G. Activation of the gut microbiota-kynurenine-liver axis contributes to the development of nonalcoholic hepatic steatosis in nondiabetic adults. *Aging* **13**, 21309–21324 (2021).
129. Arto, C. et al. Metabolic profiling of tryptophan pathways: implications for obesity and metabolic dysfunction-associated steatotic liver disease. *Eur. J. Clin. Invest.* **54**, e14279 (2024).
130. Wang, L. et al. Gut-derived serotonin contributes to the progression of non-alcoholic steatohepatitis via the liver HTR2A/PPARγ2 pathway. *Front. Pharmacol.* **11**, 553 (2020).
131. Wegermann, K. et al. Serum bile acid, vitamin E, and serotonin metabolites are associated with future liver-related events in nonalcoholic fatty liver disease. *Hepatol. Commun.* **5**, 608–617 (2021).
132. Opitz, C. A. et al. An endogenous tumour-promoting ligand of the human aryl hydrocarbon receptor. *Nature* **478**, 197–203 (2011).
133. Fujigaki, S. et al. Lipopolysaccharide induction of indoleamine 2,3-dioxygenase is mediated dominantly by an IFN-γ-independent mechanism. *Eur. J. Immunol.* **31**, 2313–2318 (2001).
134. Laurans, L. et al. Genetic deficiency of indoleamine 2,3-dioxygenase promotes gut microbiota-mediated metabolic health. *Nat. Med.* **24**, 1113–1120 (2018).
135. Choi, W. et al. Serotonin signals through a gut-liver axis to regulate hepatic steatosis. *Nat. Commun.* **9**, 4824 (2018).
136. Crane, J. D. et al. Inhibiting peripheral serotonin synthesis reduces obesity and metabolic dysfunction by promoting brown adipose tissue thermogenesis. *Nat. Med.* **21**, 166–172 (2015).
137. Gao, K., Mu, C. L., Farzi, A. & Zhu, W. Y. Tryptophan metabolism: a link between the gut microbiota and brain. *Adv. Nutr.* **11**, 709–723 (2020).
138. Rothhammer, V. et al. Type I interferons and microbial metabolites of tryptophan modulate astrocyte activity and central nervous system inflammation via the aryl hydrocarbon receptor. *Nat. Med.* **22**, 586–597 (2016).
139. Stockinger, B., Shah, K. & Wincent, E. AHR in the intestinal microenvironment: safeguarding barrier function. *Nat. Rev. Gastroenterol. Hepatol.* **18**, 559–570 (2021).
140. Ma, L. et al. Indole alleviates diet-induced hepatic steatosis and inflammation in a manner involving myeloid cell 6-phosphofructo-2-kinase/fructose-2,6-bisphosphatase 3. *Hepatology* **72**, 1191–1203 (2020).
141. Yu, J. S. et al. *Lactobacillus lactis* and *Pediococcus pentosaceus*-driven reprogramming of gut microbiome and metabolome ameliorates the progression of non-alcoholic fatty liver disease. *Clin. Transl. Med.* **11**, e634 (2021).
142. Ji, Y., Gao, Y., Chen, H., Yin, Y. & Zhang, W. Indole-3-acetic acid alleviates nonalcoholic fatty liver disease in mice via attenuation of hepatic lipogenesis, and oxidative and inflammatory stress. *Nutrients* **11**, 2062 (2019).
143. Zhao, Z. H. et al. Indole-3-propionic acid inhibits gut dysbiosis and endotoxin leakage to attenuate steatohepatitis in rats. *Exp. Mol. Med.* **51**, 1–14 (2019).
144. Zhu, B. et al. Indole supplementation ameliorates MCD-induced NASH in mice. *J. Nutr. Biochem.* **107**, 109041 (2022).
145. Mbaye, B. et al. A catalog of ethanol-producing microbes in humans. *Future Microbiol.* **19**, 697–714 (2024).
146. Yuan, J. et al. Fatty liver disease caused by high-alcohol-producing *Klebsiella pneumoniae*. *Cell Metab.* **30**, 675–688.e7 (2019).
147. Meijnikman, A. S. et al. Microbiome-derived ethanol in nonalcoholic fatty liver disease. *Nat. Med.* **28**, 2100–2106 (2022).
148. Mbaye, B. et al. Endogenous ethanol and triglyceride production by gut *Pichia kudriavzevii*, *Candida albicans* and *Candida glabrata* yeasts in non-alcoholic steatohepatitis. *Cells* **11**, 3390 (2022).
149. Seki, E. et al. TLR4 enhances TGF-β signaling and hepatic fibrosis. *Nat. Med.* **13**, 1324–1332 (2007).
150. Rivera, C. A. et al. Toll-like receptor-4 signaling and Kupffer cells play pivotal roles in the pathogenesis of non-alcoholic steatohepatitis. *J. Hepatol.* **47**, 571–579 (2007).
151. Yu, J. et al. Hepatocyte TLR4 triggers inter-hepatocyte Jagged1/Notch signaling to determine NASH-induced fibrosis. *Sci. Transl. Med.* **13**, eabe1692 (2021).
152. Swanson, K. V., Deng, M. & Ting, J. P. The NLRP3 inflammasome: molecular activation and regulation to therapeutics. *Nat. Rev. Immunol.* **19**, 477–489 (2019).
153. Csak, T. et al. Fatty acid and endotoxin activate inflammasomes in mouse hepatocytes that release danger signals to stimulate immune cells. *Hepatology* **54**, 133–144 (2011).
154. Vandanmagsar, B. et al. The NLRP3 inflammasome instigates obesity-induced inflammation and insulin resistance. *Nat. Med.* **17**, 179–188 (2011).
155. Gaul, S. et al. Hepatocyte pyroptosis and release of inflammasome particles induce stellate cell activation and liver fibrosis. *J. Hepatol.* **74**, 156–167 (2021).
156. Henaoui-Mejia, J. et al. Inflammasome-mediated dysbiosis regulates progression of NAFLD and obesity. *Nature* **482**, 179–185 (2012).
157. Pierantonelli, I. et al. Lack of NLRP3-inflammasome leads to gut-liver axis derangement, gut dysbiosis and a worsened phenotype in a mouse model of NAFLD. *Sci. Rep.* **7**, 12200 (2017).
158. Kaufmann, B. et al. Cell-specific deletion of NLRP3 inflammasome identifies myeloid cells as key drivers of liver inflammation and fibrosis in murine steatohepatitis. *Cell Mol. Gastroenterol. Hepatol.* **14**, 751–767 (2022).
159. Remetic, J. et al. Loss of bile salt export pump aggravates lipopolysaccharide-induced liver injury in mice due to impaired hepatic endotoxin clearance. *Hepatology* **75**, 1095–1109 (2022).
160. Levine, D. M., Parker, T. S., Donnelly, T. M., Walsh, A. & Rubin, A. L. In vivo protection against endotoxin by plasma high density lipoprotein. *Proc. Natl Acad. Sci. USA* **90**, 12040–12044 (1993).
161. Han, Y. H. et al. Enterically derived high-density lipoprotein restrains liver injury through the portal vein. *Science* **373**, eabe6729 (2021).

162. Nikeghbalian, S. et al. Correlation between HDL2, HDL3 and serum ferritin levels with fatty liver and NAFLD activity score (NAS) in liver histology of organ donors. *BMC Gastroenterol.* **21**, 405 (2021).
163. Jing, Y. Y. et al. Toll-like receptor 4 signaling promotes epithelial-mesenchymal transition in human hepatocellular carcinoma induced by lipopolysaccharide. *BMC Med.* **10**, 98 (2012).
164. Liu, W. T. et al. Toll like receptor 4 facilitates invasion and migration as a cancer stem cell marker in hepatocellular carcinoma. *Cancer Lett.* **358**, 136–143 (2015).
165. Dapito, D. H. et al. Promotion of hepatocellular carcinoma by the intestinal microbiota and TLR4. *Cancer Cell* **21**, 504–516 (2012).
166. Liu, W. T. et al. Lipopolysaccharide induces the differentiation of hepatic progenitor cells into myofibroblasts constitutes the hepatocarcinogenesis-associated microenvironment. *Cell Death Differ.* **27**, 85–101 (2020).
167. Ansari, S. R. et al. Risk factors for infections with multidrug-resistant *Stenotrophomonas maltophilia* in patients with cancer. *Cancer* **109**, 2615–2622 (2007).
168. Safdar, A. & Rolston, K. V. *Stenotrophomonas maltophilia*: changing spectrum of a serious bacterial pathogen in patients with cancer. *Clin. Infect. Dis.* **45**, 1602–1609 (2007).
169. Barrow, F. et al. Microbiota-driven activation of intrahepatic B cells aggravates NASH through innate and adaptive signaling. *Hepatology* **74**, 704–722 (2021).
170. Kotsiliti, E. et al. Intestinal B cells license metabolic T-cell activation in NASH microbiota/antigen-independently and contribute to fibrosis by IgA-FcR signalling. *J. Hepatol.* **79**, 296–313 (2023).
171. Miyamoto, Y. et al. Periportal macrophages protect against commensal-driven liver inflammation. *Nature* **629**, 901–909 (2024).
172. Pfister, D. et al. NASH limits anti-tumour surveillance in immunotherapy-treated HCC. *Nature* **592**, 450–456 (2021).
173. Li, M. et al. Spatial proteomics of immune microenvironment in nonalcoholic steatohepatitis-associated hepatocellular carcinoma. *Hepatology* **79**, 560–574 (2024).
174. Tincopa, M. A. & Loomba, R. Non-invasive diagnosis and monitoring of non-alcoholic fatty liver disease and non-alcoholic steatohepatitis. *Lancet Gastroenterol. Hepatol.* **8**, 660–670 (2023).
175. Guha, I. N. et al. Noninvasive markers of fibrosis in nonalcoholic fatty liver disease: validating the European liver fibrosis panel and exploring simple markers. *Hepatology* **47**, 455–460 (2008).
176. Harrison, S. A. et al. A blood-based biomarker panel (NIS4) for non-invasive diagnosis of non-alcoholic steatohepatitis and liver fibrosis: a prospective derivation and global validation study. *Lancet Gastroenterol. Hepatol.* **5**, 970–985 (2020).
177. Vali, Y. et al. Biomarkers for staging fibrosis and non-alcoholic steatohepatitis in non-alcoholic fatty liver disease (the LITMUS project): a comparative diagnostic accuracy study. *Lancet Gastroenterol. Hepatol.* **8**, 714–725 (2023).
178. Zhang, X. et al. A blood-based biomarker panel for non-invasive diagnosis of metabolic dysfunction-associated steatohepatitis. *Cell Metab.* **37**, 59–68.e3 (2025).
179. Lee, P. C. et al. Gut microbiota and metabolites associate with outcomes of immune checkpoint inhibitor-treated unresectable hepatocellular carcinoma. *J. Immunother. Cancer* **10**, e004779 (2022).
180. Ponziani, F. R. et al. Gut dysbiosis and fecal calprotectin predict response to immune checkpoint inhibitors in patients with hepatocellular carcinoma. *Hepatol. Commun.* **6**, 1492–1501 (2022).
181. Derosa, L. et al. Intestinal *Akkermansia muciniphila* predicts clinical response to PD-1 blockade in patients with advanced non-small-cell lung cancer. *Nat. Med.* **28**, 315–324 (2022).
182. Iida, N. et al. Overuse of antianaerobic drug is associated with poor postchemotherapy prognosis of patients with hepatocellular carcinoma. *Int. J. Cancer* **145**, 2701–2711 (2019).
183. Zhou, D. et al. Total fecal microbiota transplantation alleviates high-fat diet-induced steatohepatitis in mice via beneficial regulation of gut microbiota. *Sci. Rep.* **7**, 1529 (2017).
184. Xue, L., Deng, Z., Luo, W., He, X. & Chen, Y. Effect of fecal microbiota transplantation on non-alcoholic fatty liver disease: a randomized clinical trial. *Front. Cell Infect. Microbiol.* **12**, 759306 (2022).
185. Craven, L. et al. Allogenic fecal microbiota transplantation in patients with nonalcoholic fatty liver disease improves abnormal small intestinal permeability: a randomized control trial. *Am. J. Gastroenterol.* **115**, 1055–1065 (2020).
186. Witjes, J. J. et al. Donor fecal microbiota transplantation alters gut microbiota and metabolites in obese individuals with steatohepatitis. *Hepatol. Commun.* **4**, 1578–1590 (2020).
187. Stols-Goncalves, D. et al. Faecal microbiota transplantation affects liver DNA methylation in non-alcoholic fatty liver disease: a multi-omics approach. *Gut Microbes* **15**, 2223330 (2023).
188. Bajaj, J. S. et al. Association between intestinal microbiota collected at hospital admission and outcomes of patients with cirrhosis. *Clin. Gastroenterol. Hepatol.* **17**, 756–765.e3 (2019).
189. Cammarota, G. et al. European consensus conference on faecal microbiota transplantation in clinical practice. *Gut* **66**, 569–580 (2017).
190. Loguercio, C. et al. Beneficial effects of a probiotic VSL#3 on parameters of liver dysfunction in chronic liver diseases. *J. Clin. Gastroenterol.* **39**, 540–543 (2005).
191. Alisi, A. et al. Randomised clinical trial: the beneficial effects of VSL#3 in obese children with non-alcoholic steatohepatitis. *Aliment. Pharmacol. Ther.* **39**, 1276–1285 (2014).
192. Duseja, A. et al. High potency multistrain probiotic improves liver histology in non-alcoholic fatty liver disease (NAFLD): a randomised, double-blind, proof of concept study. *BMJ Open Gastroenterol.* **6**, e000315 (2019).
193. Chong, P. L., Laight, D., Aspinall, R. J., Higginson, A. & Cummings, M. H. A randomised placebo controlled trial of VSL#3 probiotic on biomarkers of cardiovascular risk and liver injury in non-alcoholic fatty liver disease. *BMC Gastroenterol.* **21**, 144 (2021).
194. Wong, V. W. et al. Treatment of nonalcoholic steatohepatitis with probiotics: a proof-of-concept study. *Ann. Hepatol.* **12**, 256–262 (2013).
195. Aller, R. et al. Effect of a probiotic on liver aminotransferases in nonalcoholic fatty liver disease patients: a double blind randomized clinical trial. *Eur. Rev. Med. Pharmacol. Sci.* **15**, 1090–1095 (2011).
196. Dao, M. C. et al. *Akkermansia muciniphila* and improved metabolic health during a dietary intervention in obesity: relationship with gut microbiome richness and ecology. *Gut* **65**, 426–436 (2016).
197. Zhu, W., Yan, M., Cao, H., Zhou, J. & Xu, Z. Effects of *Clostridium butyricum* capsules combined with rosuvastatin on intestinal flora, lipid metabolism, liver function and inflammation in NAFLD patients. *Cell Mol. Biol.* **68**, 64–69 (2022).
198. Everard, A., Matamoros, S., Geurts, L., Delzenne, N. M. & Cani, P. D. *Saccharomyces boulardii* administration changes gut microbiota and reduces hepatic steatosis, low-grade inflammation, and fat mass in obese and type 2 diabetic db/db mice. *mBio* **5**, e01011-14 (2014).
199. Yang, A. M. et al. *Saccharomyces boulardii* ameliorates non-alcoholic steatohepatitis in mice induced by a methionine-choline-deficient diet through gut-liver axis. *Front. Microbiol.* **13**, 887728 (2022).
200. Guslandi, M., Ghollo, P. & Testoni, P. A. A pilot trial of *Saccharomyces boulardii* in ulcerative colitis. *Eur. J. Gastroenterol. Hepatol.* **15**, 697–698 (2003).
201. Lang, S. & Schnabl, B. Microbiota and fatty liver disease—the known, the unknown, and the future. *Cell Host Microbe* **28**, 233–244 (2020).
202. Hendriks, T. et al. Bacteria engineered to produce IL-22 in intestine induce expression of REG3G to reduce ethanol-induced liver disease in mice. *Gut* **68**, 1504–1515 (2019).
203. Huang, X. et al. Fructooligosaccharides attenuate non-alcoholic fatty liver disease by remodeling gut microbiota and association with lipid metabolism. *Biomed. Pharmacother.* **159**, 114300 (2023).
204. Ni, Y. et al. Resistant starch decreases intrahepatic triglycerides in patients with NAFLD via gut microbiome alterations. *Cell Metab.* **35**, 1530–1547.e8 (2023).
205. Zhang, X., Yu, J. & Wong, V. W. Live la resistant starch: a potential treatment for metabolic dysfunction-associated steatohepatitis. *Cell Metab.* **35**, 1491–1493 (2023).
206. Krawczyk, M. et al. Gut permeability might be improved by dietary fiber in individuals with nonalcoholic fatty liver disease (NAFLD) undergoing weight reduction. *Nutrients* **10**, 1793 (2018).
207. Scorletti, E. et al. Synbiotics alter fecal microbiomes, but not liver fat or fibrosis, in a randomized trial of patients with nonalcoholic fatty liver disease. *Gastroenterology* **158**, 1597–1610.e7 (2020).
208. Salminen, S. et al. The International Scientific Association of Probiotics and Prebiotics (ISAPP) consensus statement on the definition and scope of postbiotics. *Nat. Rev. Gastroenterol. Hepatol.* **18**, 649–667 (2021).
209. Plovier, H. et al. A purified membrane protein from *Akkermansia muciniphila* or the pasteurized bacterium improves metabolism in obese and diabetic mice. *Nat. Med.* **23**, 107–113 (2017).
210. Depommier, C. et al. Supplementation with *Akkermansia muciniphila* in overweight and obese human volunteers: a proof-of-concept exploratory study. *Nat. Med.* **25**, 1096–1103 (2019).
211. Brandt, A. et al. Antibiotics attenuate diet-induced nonalcoholic fatty liver disease without altering intestinal barrier dysfunction. *J. Nutr. Biochem.* **123**, 109495 (2024).
212. Jian, J. et al. Rifaximin ameliorates non-alcoholic steatohepatitis in mice through regulating gut microbiome-related bile acids. *Front. Pharmacol.* **13**, 841132 (2022).
213. Abdel-Razik, A. et al. Rifaximin in nonalcoholic fatty liver disease: hit multiple targets with a single shot. *Eur. J. Gastroenterol. Hepatol.* **30**, 1237–1246 (2018).
214. Wu, P. et al. Ursodeoxycholic acid alleviates nonalcoholic fatty liver disease by inhibiting apoptosis and improving autophagy via activating AMPK. *Biochem. Biophys. Res. Commun.* **529**, 834–838 (2020).
215. Leuschner, U. F. et al. High-dose ursodeoxycholic acid therapy for nonalcoholic steatohepatitis: a double-blind, randomized, placebo-controlled trial. *Hepatology* **52**, 472–479 (2010).
216. Zhang, W., Tang, Y., Huang, J. & Hu, H. Efficacy of ursodeoxycholic acid in nonalcoholic fatty liver disease: an updated meta-analysis of randomized controlled trials. *Asia Pac. J. Clin. Nutr.* **29**, 696–705 (2020).
217. Patel, V. S. et al. Ursodeoxycholic acid's effectiveness in the management of nonalcoholic fatty liver disease: a systematic review and meta-analysis. *Euroasian J. Hepatogastroenterol.* **14**, 92–98 (2024).
218. Li, X. et al. Short-chain fatty acids in nonalcoholic fatty liver disease: new prospects for short-chain fatty acids as therapeutic targets. *Heliyon* **10**, e26991 (2024).
219. Fang, W., Xue, H., Chen, X., Chen, K. & Ling, W. Supplementation with sodium butyrate modulates the composition of the gut microbiota and ameliorates high-fat diet-induced obesity in mice. *J. Nutr.* **149**, 747–754 (2019).
220. Gan, L. et al. Bacteriophage targeting microbiota alleviates non-alcoholic fatty liver disease induced by high alcohol-producing *Klebsiella pneumoniae*. *Nat. Commun.* **14**, 3215 (2023).
221. Federici, S. et al. Targeted suppression of human IBD-associated gut microbiota commensals by phage consortia for treatment of intestinal inflammation. *Cell* **185**, 2879–2898.e24 (2022).

222. Hsu, C. L., Duan, Y., Fouts, D. E. & Schnabl, B. Intestinal virome and therapeutic potential of bacteriophages in liver disease. *J. Hepatol.* **75**, 1465–1475 (2021).
223. Hampton, H. G., Watson, B. N. J. & Fineran, P. C. The arms race between bacteria and their phage foes. *Nature* **577**, 327–336 (2020).
224. Gu, J. et al. A method for generation phage cocktail with great therapeutic potential. *PLoS ONE* **7**, e31698 (2012).
225. Kirby, A. E. Synergistic action of gentamicin and bacteriophage in a continuous culture population of *Staphylococcus aureus*. *PLoS ONE* **7**, e51017 (2012).
226. Verma, V., Harjai, K. & Chhibber, S. Restricting ciprofloxacin-induced resistant variant formation in biofilm of *Klebsiella pneumoniae* B5055 by complementary bacteriophage treatment. *J. Antimicrob. Chemother.* **64**, 1212–1218 (2009).
227. Oechslin, F. Resistance development to bacteriophages occurring during bacteriophage therapy. *Viruses* **10**, 351 (2018).
228. Shi, Y., Wang, G., Lau, H. C. & Yu, J. Metagenomic sequencing for microbial DNA in human samples: emerging technological advances. *Int. J. Mol. Sci.* **23**, 2181 (2022).
229. Meslier, V. et al. Benchmarking second and third-generation sequencing platforms for microbial metagenomics. *Sci. Data* **9**, 694 (2022).
230. Pereira, M. B., Wallroth, M., Jonsson, V. & Kristiansson, E. Comparison of normalization methods for the analysis of metagenomic gene abundance data. *BMC Genom.* **19**, 274 (2018).
231. Lucchetti, M. et al. An organ-on-chip platform for simulating drug metabolism along the gut-liver Axis. *Adv. Healthc. Mater.* **13**, e2303943 (2024).
232. Sun, Y. et al. Integrative plasma and fecal metabolomics identify functional metabolites in adenoma-colorectal cancer progression and as early diagnostic biomarkers. *Cancer Cell* **42**, 1386–1400.e8 (2024).
233. Asnicar, F. et al. Blue poo: impact of gut transit time on the gut microbiome using a novel marker. *Gut* **70**, 1665–1674 (2021).
234. Roager, H. M. et al. Colonic transit time is related to bacterial metabolism and mucosal turnover in the gut. *Nat. Microbiol.* **1**, 16093 (2016).
235. Falony, G. et al. Population-level analysis of gut microbiome variation. *Science* **352**, 560–564 (2016).
236. Prochazkova, N. et al. Advancing human gut microbiota research by considering gut transit time. *Gut* **72**, 180–191 (2023).
237. Raba, G., Adamberg, S. & Adamberg, K. Acidic pH enhances butyrate production from pectin by faecal microbiota. *FEMS Microbiol. Lett.* **368**, fnab042 (2021).
238. Ting, N. L., Lau, H. C. & Yu, J. Cancer pharmacomicrobiomics: targeting microbiota to optimise cancer therapy outcomes. *Gut* **71**, 1412–1425 (2022).
239. Yang, Y. et al. Fecal microbiota transplantation: no longer cinderella in tumour immunotherapy. *EBioMedicine* **100**, 104967 (2024).
240. Fong, W. et al. *Lactobacillus gallinarum*-derived metabolites boost anti-PD1 efficacy in colorectal cancer by inhibiting regulatory T cells through modulating IDO1/Kyn/AHR axis. *Gut* **72**, 2272–2285 (2023).
241. Del Chierico, F. et al. Gut microbiota profiling of pediatric nonalcoholic fatty liver disease and obese patients unveiled by an integrated meta-omics-based approach. *Hepatology* **65**, 451–464 (2017).
242. Boursier, J. et al. The severity of nonalcoholic fatty liver disease is associated with gut dysbiosis and shift in the metabolic function of the gut microbiota. *Hepatology* **63**, 764–775 (2016).
243. Ratzl, V. et al. A randomized controlled trial of high-dose ursodesoxycholic acid for nonalcoholic steatohepatitis. *J. Hepatol.* **54**, 1011–1019 (2011).
244. Rinella, M. E. et al. A multisociety Delphi consensus statement on new fatty liver disease nomenclature. *J. Hepatol.* **79**, 1542–1556 (2023).
245. Younossi, Z. M. et al. The global epidemiology of NAFLD and NASH in patients with type 2 diabetes: a systematic review and meta-analysis. *J. Hepatol.* **71**, 793–801 (2019).
246. Younossi, Z. M. et al. The global epidemiology of nonalcoholic fatty liver disease (NAFLD) and nonalcoholic steatohepatitis (NASH): a systematic review. *Hepatology* **77**, 1335–1347 (2023).
247. Huang, D. Q., El-Serag, H. B. & Loomba, R. Global epidemiology of NAFLD-related HCC: trends, predictions, risk factors and prevention. *Nat. Rev. Gastroenterol. Hepatol.* **18**, 223–238 (2021).
248. The Lancet Gastroenterology Hepatology. Resmetirom for NASH: balancing promise and prudence. *Lancet Gastroenterol. Hepatol.* **9**, 273 (2024).
249. Newsome, P. N. et al. A placebo-controlled trial of subcutaneous semaglutide in nonalcoholic steatohepatitis. *N. Engl. J. Med.* **384**, 1113–1124 (2021).
250. Loomba, R. et al. Randomized, controlled trial of the FGF21 analogue pegozafermin in NASH. *N. Engl. J. Med.* **389**, 998–1008 (2023).
251. Harrison, S. A. et al. Safety and efficacy of once-weekly efruxifermin versus placebo in non-alcoholic steatohepatitis (HARMONY): a multicentre, randomised, double-blind, placebo-controlled, phase 2b trial. *Lancet Gastroenterol. Hepatol.* **8**, 1080–1093 (2023).
252. Francke, S. M. et al. A randomized, controlled trial of the Pan-PPAR agonist lanifibranor in NASH. *N. Engl. J. Med.* **385**, 1547–1558 (2021).

Acknowledgements

The authors were supported by the Noncommunicable Chronic Diseases-National Science and Technology Major Project (2023ZD0500200), the National Natural Science Foundation of China (82222901 and 82272619) and the Research Grants Council-General Research Fund (CUHK14117422, CUHK14119524 and CUHK14117123).

Author contributions

H.C.-H.L. and X.Z. researched data for the article and wrote the manuscript. J.Y. made a substantial contribution to discussion of content and reviewed/edited the manuscript before submission.

Competing interests

The authors declare no competing interests.

Additional information

Peer review information *Nature Reviews Gastroenterology & Hepatology* thanks Münevver Demir and the other, anonymous, reviewer(s) for their contribution to the peer review of this work.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

© Springer Nature Limited 2025